Fluid Dynamics, Orders and Quantum Information

Microscopic reduction, finite complexes, quantum learning and a structural reading of the 2026 Navier–Stokes release

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A standalone extension of the research monograph

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Purpose and scope

This supplement develops the fluid-dynamics extension of *Orders, Complexes, Cryptography and Quantum Yang–Mills*. It contains the same four fluid chapters and verification appendix as the complete monograph, with independent pagination. The bibliography identifies the primary sources, and the companion repository contains the source coverage, exact formal declarations, numerical programs and interactive laboratory.

The contribution is an explicit mathematical interpretation and a collection of proved interfaces. The released continuum singularity theorem is attributed to its source. The selected local Lean declarations and finite computations have their own verification records. We do not present a stronger unforced theorem, a fully formalized version of every chapter, or a trained fluid model where none was executed.

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Chapter 1

Microscopic information, kinetic limits, and fluid closure

Evidence status. The 249 timed English transcript segments of the requested braintruffle video, from 0:00 through 32:36, were reviewed in the public YouTube transcript interface. The timestamped source audit records precisely which accompanying papers, lecture decks, and book sections were accessible. The full main texts of Stodolna, Vozzi, Itatani, Allen, the abridged English Jönsson article, and all six requested Volkov decks were reviewed. Several other requested originals were accessible only as abstracts or metadata; these are not represented as full-text reviews. The arguments proved below are ordinary mathematical proofs. Their presence here does not assert a Lean formalization of the continuum theories.

1.1 What the microscopic perspective establishes

The video organizes fluid simulation around successive reductions in the information retained about a physical system: a quantum state, molecular positions and velocities, a velocity distribution, and a few macroscopic fields [5]. This is a productive organizing principle provided that each reduction is supplied with its own assumptions. An exact change of coordinates, a statistical marginal, an asymptotic limit, and an empirical constitutive law are mathematically different operations. The distinction determines which conclusions can be carried from one level to another.

For example, expanding a wave function in a complete eigenbasis retains all information. Keeping only finitely many modes discards information. Taking the partial trace of a many-body density operator discards correlations. Replacing a quantum nuclear state by a classical trajectory requires a localization or semiclassical approximation. Replacing an N-particle classical distribution by a one-particle distribution requires a closure of its correlation hierarchy. Replacing that distribution by density, velocity, and temperature requires another closure. None of these steps follows merely from a count of variables.

The central mathematical lesson is therefore a sequence of conditional statements. The Born–Huang expansion identifies the terms discarded by an adiabatic approximation. Liouville's equation identifies the correlations discarded by a kinetic approximation. Collision invariants identify the moments retained by a fluid approximation. An entropy inequality restricts admissible closures but does not determine all their coefficients. A hydrodynamic-limit theorem supplies compactness and convergence information that a formal moment calculation does not supply. The structural results in Chapter 3 and the quantum algorithms in Chapter 4 fit into these particular stages.

1.1.1 Five distinct mathematical state spaces

For N distinguishable spinless particles in three dimensions, a pure quantum state is a unit vector in $L^2(\mathbb{R}^{3N})$, up to an overall phase. An N-particle classical microstate is a point of $\mathbb{R}^{6N}$. An uncertain classical state is instead a probability measure on that same phase space; when it has a density, that density is a function of $6N$ variables. Thus the video's contrast between a wave function of $3N$ coordinates and a classical list of $6N$ numbers compares a general quantum state with a single classical microstate. A comparison of statistical descriptions must also include classical ensemble distributions.

The one-particle kinetic density $f(t, x, v)$ is a function on $\mathbb{R}^3_x \times \mathbb{R}^3_v$. In the unit-mass convention used below, its basic moments are

$$\rho = \int f \, dv, \qquad \rho u = \int v f \, dv, \qquad 3\rho\theta = \int |v - u|^2 f \, dv. \tag{1.1}$$

Here $\theta = k_B T/m$ in physical units. The fluid variables (ρ, u, θ) contain five scalar fields. They generally do not determine f, its pressure tensor, or its heat flux.

In the quantum setting, mixed states are positive trace-one operators ϱ. A reduced one-particle density matrix and a classical kinetic density are different kinds of objects. A Wigner transform can relate them in suitable semiclassical regimes, but a generic Wigner function is not a nonnegative probability density. A classical probabilistic interpretation therefore cannot be inserted without a positivity or limiting argument.

1.2 Measurement, phase, and the limits of tomography

The experiments cited by the video support wave interference and model-dependent reconstruction of electronic structure. Jönsson's multi-slit experiment compares electron diffraction with interference predictions [18]. Stodolna and collaborators image nodal structures in an outgoing electron distribution associated with hydrogen Stark states; the detector distribution is related to a propagated wave by the apparatus and its model [29]. Itatani's molecular-orbital tomography uses high-harmonic emission, alignment information, a reference atom, and a recollision approximation [15]. Vozzi's generalized tomography improves phase recovery and the treatment of Coulomb distortion while retaining finite-bandwidth and reconstruction assumptions [39].

These are not measurements of a particle trajectory or unrestricted tomography of an arbitrary many-electron wave function. In particular, a single orbital reconstruction does not by itself determine every many-particle correlation. The distinction can be expressed without depending on an interpretation of quantum mechanics.

Proposition 1.1 (Finite-dimensional informational completeness)**.** *Let $E_1, \dots, E_s$ be positive operators on $\mathbb{C}^d$ satisfying $\sum_i E_i = I$. The probabilities $p_i = \operatorname{Tr}(E_i \varrho)$ uniquely determine every density operator ϱ if and only if the real linear span of the E_i is the space of Hermitian $d \times d$ matrices.*

Proof. If the span is the full Hermitian space and two states give the same probabilities, their difference X is orthogonal to every Hermitian matrix in the inner product $\operatorname{Tr}(XY)$. Taking $Y = X$ gives $\operatorname{Tr}(X^2) = 0$, hence $X = 0$.

Conversely, if the span is proper, choose a nonzero Hermitian matrix X orthogonal to it. Since $I = \sum_i E_i$ belongs to the span, $\operatorname{Tr} X = 0$. For sufficiently small $a > 0$, both $\varrho_\pm = I/d \pm aX$ are positive trace-one matrices. They are distinct and satisfy $\operatorname{Tr}(E_i \varrho_+) = \operatorname{Tr}(E_i \varrho_-)$ for every i. □

Position probabilities alone fail this criterion. The qubit vectors $(|0\rangle+|1\rangle)/\sqrt{2}$ and $(|0\rangle-|1\rangle)/\sqrt{2}$ have identical computational-basis probabilities and different relative phase. Similarly, $\psi(x)$ and $e^{i\alpha(x)}\psi(x)$ have the same $|\psi(x)|^2$ where both define admissible states, but may have different currents and future dynamics. The Born density for a complex wave function is $|\psi|^2$, not its algebraic square.

Proposition 1.2 (Local operations do not transmit a signal through a marginal)**.** *Let ϱ_{AB} be a density operator on a finite tensor product. For operators K_j on B satisfying $\sum_j K_j^* K_j = I_B$, define*

$$\widetilde{\varrho}_{AB} = \sum_j (I_A \otimes K_j)\varrho_{AB}(I_A \otimes K_j^*).$$

Then $\mathrm{Tr}_B\, \widetilde{\varrho}_{AB} = \mathrm{Tr}_B\, \varrho_{AB}$.

Proof. For every observable A on the first factor, cyclicity of the trace gives

$$\mathrm{Tr}((A\otimes I)\widetilde{\varrho}) = \sum_j \mathrm{Tr}((A\otimes K_j^* K_j)\varrho) = \mathrm{Tr}((A\otimes I)\varrho).$$

Equality against every A determines the reduced operator. □

Conditioning on a recorded outcome can change a conditional state. The outcome record must then be communicated before the other observer can condition on it. Bell correlations consequently do not justify an assertion of controllable instantaneous communication. This qualification is needed when using the introductory discussion in van Dommelen's book [32].

1.2.1 What the multi-slit formula proves

In the Fraunhofer approximation, suppose s equally illuminated identical slits contribute a common amplitude $a(\vartheta)$ and a phase increment $\delta = 2\pi d \sin\vartheta/\lambda$ between neighboring slits. Linearity gives

$$A(\vartheta) = a(\vartheta)\sum_{j=0}^{s-1} e^{ij\delta}.$$

The geometric-series identity and continuity at removable singularities give

$$|A(\vartheta)|^2 = |a(\vartheta)|^2 \frac{\sin^2(s\delta/2)}{\sin^2(\delta/2)}. \tag{1.2}$$

For a uniformly illuminated slit of width b, the Fourier integral $\int_{-b/2}^{b/2} e^{2\pi i x \sin\vartheta/\lambda}\, dx$ supplies a squared sinc envelope. Equation (1.2) therefore proves the interference factor under stated coherence, illumination, and far-field assumptions. It does not remove the need to model the source, finite detector resolution, or partial coherence. The reviewed 1974 article explicitly identifies itself as an abridged translation; it cannot stand in for a complete review of the longer 1961 German original [17, 18].

1.3 Exterior powers are a physical state constraint

The project's exterior algebra is directly relevant to identical fermions. Let V be a one-particle space including spin. The antisymmetric N-particle subspace is $\bigwedge^N V$, identified with the antisymmetric tensors inside $V^{\otimes N}$. Exchanging two complete spin-space labels changes a fermionic state by a sign. Equality of position probabilities under an exchange, by itself, does not specify that sign or the spin dependence.

Proposition 1.3 (Slater normalization and dimension)**.** *If $e_1, \dots, e_M$ are an orthonormal basis of the finite-dimensional one-particle space V, then for $1 \le i_1 < \cdots < i_N \le M$ the tensors*

$$\Psi_{i_1 \cdots i_N} = \frac{1}{\sqrt{N!}} \sum_{\sigma \in S_N} \operatorname{sgn}(\sigma) e_{i_{\sigma(1)}} \otimes \cdots \otimes e_{i_{\sigma(N)}}$$

form an orthonormal basis of $\bigwedge^N V$. In particular, $\dim \bigwedge^N \mathbb{C}^M = \binom{M}{N}$*, with the dimension zero for $N > M$.*

Proof. For fixed distinct indices the $N!$ elementary tensors in the sum are mutually orthonormal, so the squared norm is $N!/N! = 1$. Different index sets have disjoint elementary-tensor supports. Antisymmetry forces the coefficient of any tensor with a repeated index to vanish, and relates all coefficients belonging to permutations of a distinct index set by the permutation sign. Thus every antisymmetric tensor is a linear combination of the displayed vectors. □

In position representation this is the normalized Slater determinant. The sign representation of the symmetric group selects the Young diagram (1^N), while symmetric bosonic tensors select (N). More general Schur functors organize symmetry types of tensor features and multi-index responses. This is a precise use of the algebraic material developed earlier in the monograph: a representation specifies allowed states or tensor components. It does not, on its own, furnish a bound for a nonlinear fluid equation.

1.3.1 Number states and coherent states

Mathur's accessible introductory modules are useful for interference, tensor products, and the distinction between classical amplitude and measurement probabilities [23]. One teaching identification needs correction: a state with a fixed number of identical bosons in one mode is a number state, and is not a coherent state. Let $a|n\rangle = \sqrt{n}\,|n-1\rangle$. A coherent state is

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \tag{1.3}$$

The exponential series shows that its squared norm is one. Shifting the summation index shows $a|\alpha\rangle = \alpha|\alpha\rangle$. Its number probabilities are $e^{-|\alpha|^2}|\alpha|^{2n}/n!$, whose mean and variance are both $|\alpha|^2$. A fixed $|n\rangle$ has number variance zero and $\langle a \rangle = 0$. Quantization of the number observable therefore does not discretize every complex amplitude or every field expectation. This matters when deciding what a quantum neural model's input amplitudes and measured outputs represent.

1.4 Born–Huang expansion before Born–Oppenheimer truncation

The video's move from electronic wave functions to molecular potential energy surfaces is best made precise by first writing the coupling terms that are later neglected. The treatment in van Dommelen's Section 9.2 and derivation D.51 distinguishes electronic and nuclear quantum dynamics; Allen's molecular-dynamics notes then describe classical motion on a chosen potential [2, 32].

For an exact, domain-transparent derivation, take a finite electronic space $\mathbb{C}^r$, nuclear coordinate $R \in \mathbb{R}^d$, and a C^2 family of Hermitian matrices $H_e(R)$. Suppose an orthonormal eigenbasis $\phi_n(R)$ can be chosen C^2 on the open region being considered. For C^2 nuclear coefficient functions χ_n, set

$$H = -\frac{\hbar^2}{2M}\Delta_R + H_e(R), \qquad \Psi(R) = \sum_{n=1}^{r} \chi_n(R)\phi_n(R).$$

Scalar nuclear repulsion can be included in H_e. The corresponding continuum-electronic formulas require common operator domains and the regularity needed to justify these derivatives; the finite statement does not assert those analytic hypotheses automatically.

Proposition 1.4 (Exact coupled nuclear equations)**.** *Set*

$$d_{mn} = \langle \phi_m, \nabla_R \phi_n \rangle,$$
$$\tau_{mn} = \langle \phi_m, \Delta_R \phi_n \rangle.$$

Then $H\Psi = E\Psi$ is equivalent to

$$-\frac{\hbar^2}{2M}\left(\Delta\chi_m + 2\sum_n d_{mn}\cdot\nabla\chi_n + \sum_n \tau_{mn}\chi_n\right) + E_m(R)\chi_m = E\chi_m \tag{1.4}$$

for each m. Moreover, $d_{mn} = -\overline{d_{nm}}$ componentwise.

Proof. Apply the product rule to each $\Delta(\chi_n\phi_n)$ and take its electronic inner product with ϕ_m. Orthonormality gives the first term and diagonalizes H_e. Differentiating $\langle \phi_m, \phi_n \rangle = \delta_{mn}$ proves the final identity. □

The usual single-surface approximation retains one electronic sector and controls or omits its couplings to others. Which diagonal geometric terms are retained must be specified. Rephasing $\phi_n(R) \mapsto e^{i\gamma_n(R)}\phi_n(R)$ also rephases χ_n and changes diagonal derivative couplings; a claim that they vanish everywhere needs a gauge and topology argument.

Proposition 1.5 (Energy gradients and the gap denominator)**.** *Let $E_n(R)$ be a simple normalized eigenpair of the finite matrix family. For a coordinate derivative ∂_j,*

$$\partial_j E_n = \langle \phi_n, (\partial_j H_e)\phi_n \rangle, \qquad \langle \phi_m, \partial_j \phi_n \rangle = \frac{\langle \phi_m, (\partial_j H_e)\phi_n \rangle}{E_n - E_m} \quad (m \neq n).$$

If $|E_n - E_m| \geq \gamma > 0$ for all $m \neq n$, then

$$\|(I - |\phi_n\rangle\langle\phi_n|)\partial_j\phi_n\| \leq \gamma^{-1}\|\partial_j H_e\|.$$

Proof. Differentiate $H_e\phi_n = E_n\phi_n$ and take inner products with ϕ_n and with ϕ_m. The terms containing $\partial_j\phi_n$ cancel in the first case. In the second they give the difference $E_n - E_m$. Squaring the off-diagonal coefficients and summing over the orthonormal eigenbasis gives the stated norm bound. □

Large nuclear mass reduces the prefactor in (1.4); it does not prevent a small electronic gap from amplifying the derivative coupling. Near crossings and conical intersections, a single-surface argument requires additional work. Likewise, an excited electronic surface may have an adiabatic nuclear description. The Born–Oppenheimer approximation is not a theorem that electrons automatically dissipate into their ground state.

Proposition 1.6 (Unitary evolution does not supply ground-state relaxation)**.** *For a time-independent finite Hermitian matrix H and a spectral projector P, the population*

$$\mathrm{Tr}(Pe^{-itH/\hbar}\varrho_0 e^{itH/\hbar})$$

is constant.

Proof. The projector commutes with both exponentials. Cyclicity of the trace reduces the expression to $\mathrm{Tr}(P\varrho_0)$. □

A bath, emitted radiation, collisions with other degrees of freedom, or a specified open-system approximation can generate relaxation. Adiabatic following and relaxation are distinct physical mechanisms. The video at 21:37–21:53 compresses them into a single explanation; equations (1.4) and Proposition 1.6 state what must be separated.

1.5 When a quantum packet behaves approximately as a particle

For $H = P^2/(2M) + V(X)$, the formal Ehrenfest identities are

$$\frac{d}{dt}\langle X\rangle = \frac{\langle P\rangle}{M}, \qquad \frac{d}{dt}\langle P\rangle = -\langle \nabla V(X)\rangle. \tag{1.5}$$

They follow from $d\langle A\rangle/dt = (i/\hbar)\langle [H, A]\rangle$ on a common invariant domain. For instance, they hold for sufficiently regular rapidly decaying solutions for which all integrations by parts and differentiated moments are justified. They are not yet Newton's equations for the mean, because $\langle \nabla V(X)\rangle$ need not equal $\nabla V(\langle X\rangle)$.

Proposition 1.7 (Localization controls the mean-force error)**.** *If ∇V is L-Lipschitz and the position distribution has a finite second moment, then*

$$|\langle \nabla V(X)\rangle - \nabla V(\langle X\rangle)| \le L \left\langle |X - \langle X\rangle|^2 \right\rangle^{1/2}.$$

Proof. Apply the Lipschitz inequality inside the position integral, then the triangle inequality and Cauchy–Schwarz for its probability measure. □

This gives a quantitative condition rather than a universal particle-size threshold. To turn it into a trajectory approximation on a time interval, one must control the packet width throughout that interval and estimate the resulting perturbation of the classical flow.

For a free particle in one dimension with $\sigma > 0$ and

$$\psi_0(x) = (2\pi\sigma^2)^{-1/4} e^{-x^2/(4\sigma^2)},$$

the exact Heisenberg identity $X(t) = X + tP/M$ gives

$$\mathrm{Var}(X(t)) = \sigma^2 + \frac{\hbar^2 t^2}{4M^2\sigma^2}. \tag{1.6}$$

Indeed $\mathrm{Var}\, X = \sigma^2$, $\mathrm{Var}\, P = \hbar^2/(4\sigma^2)$, and the initial symmetrized covariance vanishes. Increasing M slows free spreading for a fixed packet, but (1.6) still grows with time. Heavy particles can also exhibit interference. A suitable scale comparison involves action, wavelength, temperature, environmental coupling, and the observable being approximated.

1.5.1 One-dimensional Coulomb models require a domain

The video explicitly uses a softened one-dimensional potential to explain electronic forces. Such a model is pedagogically useful, but it is a different operator from the singular expression

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} - \frac{\kappa}{|x|}.$$

A differential expression does not specify a self-adjoint Hamiltonian until its domain and behavior at the singular point are given. The reviewed operator-domain analysis by de Oliveira and Verri finds a family of self-adjoint realizations for the one-dimensional operator initially defined away from the origin [8]. We use that domain distinction, not an unqualified assertion about the spectrum of every regularization. The requested Loudon papers are separately registered as unavailable full texts in this review [20, 21].

The limit of a family of softened potentials must specify the topology of operator convergence and any renormalization of energy. It cannot be inferred merely by drawing potentials that approach $-\kappa/|x|$ away from zero. This is the same structural issue that recurs in continuum limits of graph Laplacians: a formula for coefficients must be accompanied by domains, boundary conditions, and convergence estimates.

1.6 Molecular dynamics and conserved microscopic information

A classical molecular Hamiltonian has the form

$$H(q,p)=\sum_{i=1}^{N}\frac{|p_i|^2}{2m_i}+U(q),\qquad \dot q_i=\frac{p_i}{m_i},\quad \dot p_i=-\nabla_{q_i}U. \tag{1.7}$$

The force is the negative gradient of potential energy. General molecular force fields include angular, torsional, many-body, constrained, or reactive terms; they need not be sums of pairwise distance functions. Allen's notes explain these modeling choices, symplectic integrators, constraints, thermostats, and statistical errors [2]. Their scope is much wider than the two-particle illustration in the video.

Proposition 1.8 (Liouville volume and fine-grained entropy)**.** *Let $H\in C^2(\mathbb{R}^{2n})$ have a smooth Hamiltonian flow Φ_t on the interval considered. Then* $\det D\Phi_t=1$*. If a smooth probability density evolves by $f_t=f_0\circ\Phi_{-t}$ and the entropy integral is absolutely integrable, then*

$$\int f_t\log f_t\,dq\,dp=\int f_0\log f_0\,dq\,dp.$$

Proof. For the Hamiltonian vector field $X_H=(\partial_pH,-\partial_qH)$, equality of mixed derivatives gives $\operatorname{div}X_H=\sum_i(\partial_{q_i}\partial_{p_i}H-\partial_{p_i}\partial_{q_i}H)=0$. Jacobi's determinant identity for the flow derivative yields $\frac{d}{dt}\det D\Phi_t=(\operatorname{div}X_H)\circ\Phi_t\det D\Phi_t=0$. Its initial value is one. A change of variables in the entropy integral proves the second assertion. □

Thus a Hamiltonian description preserves fine-grained information. Entropy growth in a reduced description is not the destruction of information by an invertible microscopic flow. It involves a marginalization, an environment, a limiting operation, or a restriction on what is resolved. Quantum unitary conjugation similarly preserves the spectrum of a density operator and hence its von Neumann entropy when that entropy is defined. A reduced subsystem can have changing entropy because its correlations with the omitted subsystem change.

1.6.1 A useful exact statement about velocity Verlet

For separable $H=T(p)+U(q)$, the kick $(q,p)\mapsto(q,p-h\nabla U(q))$ and drift $(q,p)\mapsto(q+h\nabla T(p),p)$ are exact Hamiltonian subflows. Their symmetric composition, kick by $h/2$, drift by h, kick by $h/2$, is symplectic. This proves preservation of the symplectic form; it does not prove exact preservation of the original H.

For the scalar oscillator $H = (p^2 + \omega^2 q^2)/2$, the update is

$$q' = \left(1 - \frac{h^2\omega^2}{2}\right) q + hp, \quad p' = \left(1 - \frac{h^2\omega^2}{2}\right) p - h\omega^2 \left(1 - \frac{h^2\omega^2}{4}\right) q.$$

Direct expansion proves that it preserves

$$\widetilde{H}_h(q,p) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 \left(1 - \frac{h^2\omega^2}{4}\right) q^2. \tag{1.8}$$

This quadratic form is positive definite when $0 < h\omega < 2$. It is a valid special-case modified invariant. In a general nonlinear system, a formal or asymptotic modified Hamiltonian from backward error analysis must not be promoted to an exact global invariant without the additional theorem needed for that claim.

1.7 Marginals, correlations, and the kinetic approximation

For a smooth symmetric N-particle density F_N, let $F_N^{(s)}$ denote its s-particle marginal. With unit masses and a pair force $K(x) = -\nabla\Phi(x)$, integrating the Liouville equation over all but particle 1 gives

$$\partial_t F_N^{(1)} + v \cdot \nabla_x F_N^{(1)} + (N-1)\nabla_v \cdot \int K(x-y) F_N^{(2)}(x,v,y,w)\, dy\, dw = 0, \tag{1.9}$$

assuming the boundary terms vanish. The two-particle marginal appears because the force on a particle depends on others. Its own equation involves the three-particle marginal. This is the BBGKY hierarchy.

Example 1.9 (A first marginal cannot determine a pair correlation)**.** On $\{0,1\}^2$, let P give mass $1/2$ each to $(0,0)$ and $(1,1)$, and let Q give mass $1/2$ each to $(0,1)$ and $(1,0)$. Both first and second one-coordinate marginals are uniform. Nevertheless, $P(X = Y) = 1$ and $Q(X = Y) = 0$. Thus no function of the one-coordinate marginals alone can reconstruct all pair observables.

Replacing $F_N^{(2)}$ by a product is a substantive assumption. For dilute collisions it is the incoming, appropriately scaled correlations that matter. Hard-sphere exclusion already prevents exact independence at fixed particle diameter. The word "chaos" in propagation of chaos denotes an asymptotic factorization property of fixed-order marginals; it is not interchangeable with deterministic sensitivity to initial conditions.

1.7.1 A rigorous classical-to-kinetic bridge and its time interval

In the hard-sphere Boltzmann–Grad scaling in dimension $d \geq 2$,

$$N\varepsilon^{d-1} = 1, \qquad N\varepsilon^d = \varepsilon \longrightarrow 0.$$

The first relation retains an order-one collision scale, while the second sends the occupied volume fraction to zero. Gallagher, Saint-Raymond, and Texier give a detailed derivation for hard spheres and suitable short-range repulsive potentials [12]. Their hard-sphere theorem propagates admissible asymptotic initial factorization on a positive short time interval, with convergence of velocity-averaged observables locally away from positional diagonals. It uses weighted uniform bounds on the hierarchy and quantitative control of recollisions.

The reviewed source's introductory statement assumes a continuous probability density f_0 with a Gaussian velocity bound

$$\sup_{x,v} f_0(x,v)e^{\beta|v|^2/2} < \infty \quad \text{for some } \beta > 0.$$

Its precise Theorem 8 includes the admissibility and hierarchy-space hypotheses. The proof does not assert convergence for arbitrary finite-N initial data, arbitrarily long times, every potential, or every boundary geometry. This chapter reviewed the introduction, main convergence statement, and concluding scope discussion; it does not claim to reproduce or independently verify the monograph's complete recollision proof.

This time restriction has a practical logical consequence. A short-time particle-to-Boltzmann theorem and a long macroscopic-time Boltzmann-to-fluid theorem cannot simply be concatenated. The former must hold uniformly on the time scales required by the latter, or a separate joint-limit theorem must be supplied. No amount of formal diagram composition replaces that analytic requirement.

1.8 The collision algebra and its five invariants

We now prove the algebraic facts behind the simplest elastic collision operator. They are useful both for conservative simulation and for checking learned collision models. Set $g = v - v_*$, let $n \in S^{d-1}$, and define

$$v' = v - (g \cdot n)n, \qquad v'_* = v_* + (g \cdot n)n. \tag{1.10}$$

Proposition 1.10 (Elastic reflection)**.** *The map* (1.10) *is an involution with absolute Jacobian determinant one. It preserves total momentum and kinetic energy:*

$$v' + v'_* = v + v_*, \qquad |v'|^2 + |v'_*|^2 = |v|^2 + |v_*|^2.$$

Proof. The center velocity $V = (v + v_*)/2$ is unchanged and $g' = g - 2(g \cdot n)n$. This is an orthogonal reflection, so $|g'| = |g|$, its square is the identity, and its determinant has absolute value one. Since $|v|^2 + |v_*|^2 = 2|V|^2 + |g|^2/2$, all assertions follow after the invertible linear change from (v, v_*) to (V, g). □

A scalar function ϕ is a collision invariant when $\phi(v) + \phi(v_*) = \phi(v') + \phi(v'_*)$ for every elastic collision. The functions $1, v_1, \dots, v_d, |v|^2$ are immediately invariant. The following proof makes the converse precise under a simple regularity assumption.

Proposition 1.11 (Classification of smooth collision invariants)**.** *For $d \geq 2$, every C^2 collision invariant has the form*

$$\phi(v) = a + b \cdot v + c|v|^2.$$

Consequently the scalar invariant space has dimension $d + 2$, hence five in three dimensions.

Proof. For fixed V and radius r, collisions can send g to any vector of the same length: reflection in a suitable hyperplane does so for two given vectors. Hence $\phi(V + r\omega) + \phi(V - r\omega)$ is independent of the unit vector ω. Its second-order expansion at $r = 0$ implies that $\omega^T D^2\phi(V)\omega$ is constant on the unit sphere. Taking coordinate vectors and their pairwise sums shows $D^2\phi(V) = \lambda(V)I$.

The mixed second derivatives are zero, so $\partial_i\phi(v)$ depends only on v_i. The equality of all diagonal second derivatives then says that a function of v_i alone equals a function of v_j alone for independently variable v_i, v_j. Since $d \geq 2$, each is the same constant. Integrating twice yields the displayed quadratic polynomial. The converse follows from Proposition 1.10. □

Some introductory notes group the invariants into three physical categories: particle number, momentum, and energy. Momentum is a vector, so those categories contain five independent scalar invariants in three dimensions, not three scalar equations.

1.8.1 A collision kernel contains more information than a diameter

For a central classical interaction, the differential cross section is obtained from the map between impact parameter b and scattering angle χ. On a branch where the map is invertible,

$$\frac{d\sigma}{d\Omega} = \frac{b}{\sin\chi}\left|\frac{db}{d\chi}\right|.$$

If multiple branches produce the same angle, their contributions must be added. Angular singularities and grazing collisions require further analysis. The hard-sphere model and the variable hard-sphere model used in Volkov's notes do not reproduce every angular distribution of a general potential [34]. Matching a viscosity law or a mixing statistic is a calibration of particular observables, not an equality of the underlying scattering laws.

Internal excitation and chemical reactions add variables, transitions, and conservation relations. Parker's requested paper concerns rotational and vibrational relaxation, while Valentini and collaborators' requested study concerns nitrogen dissociation on an ab initio potential surface [28, 31]. Only their accessible abstracts and metadata were reviewed here. Their reported models must not be treated as fully audited derivations or as evidence that every gas can be represented by an elastic monatomic kernel.

1.9 The weak collision identity and entropy production

Let $B(g,n) \geq 0$ be a collision kernel invariant under exchange of the particles and under the pre/post-collision reflection. Define

$$Q(f,f)(v) = \int_{\mathbb{R}^d}\int_{S^{d-1}} B(v-v_*,n)(f'f'_* - ff_*)\,dn\,dv_*. \tag{1.11}$$

Here all four values of f have the same (t,x), when those variables are present. The next statements assume absolute integrability of the displayed integrals, so that Fubini's theorem and the changes of variables are legitimate.

Proposition 1.12 (Weak collision identity)**.** *For an admissible test function ϕ,*

$$\begin{aligned}\int \phi Q(f,f)\,dv &= \frac{1}{4}\iiint B(f'f'_* - ff_*)(\phi+\phi_*-\phi'-\phi'_*)\,dn\,dv_*\,dv \\ &= \frac{1}{2}\iiint Bff_*(\phi'+\phi'_*-\phi-\phi_*)\,dn\,dv_*\,dv.\end{aligned} \tag{1.12}$$

Proof. Exchange v and v_* in the original integral and average the two expressions. In a second copy, change variables from (v,v_*) to (v',v'_*). The measure and B are unchanged, the gain-minus-loss factor changes sign, and the test functions acquire primes. Averaging once more gives the first identity. Applying the same pre/post change to the gain term gives the second. □

Testing with collision invariants proves the mass, momentum, and energy collision balances. Testing with the logarithm supplies a stronger constraint, subject to the additional integrability needed for that unbounded test.

Theorem 1.13 (Classical homogeneous H theorem under explicit regularity)**.** *Let $f(t,v) > 0$ be a differentiable solution of $\partial_t f = Q(f,f)$ for which $\int f \log f$ is finite, differentiation under the integral is valid, and all terms in (1.12) with $\phi = \log f$ are integrable. Then*

$$\frac{d}{dt}\int f \log f \, dv = -D(f) \le 0, \quad D(f) = \frac{1}{4}\iiint B(f'f'_* - ff_*) \log \frac{f'f'_*}{ff_*} \, dn \, dv_* \, dv. \tag{1.13}$$

Proof. The derivative equals $\int (1+\log f) Q(f,f)\, dv$. The constant part vanishes by mass conservation. Use (1.12) for the remaining part. Since $(a-b)(\log a - \log b) \ge 0$ for $a, b > 0$, the displayed entropy production is nonnegative. □

For spatially varying smooth solutions, transport adds the entropy flux $\nabla_x \cdot \int v f \log f \, dv$. Boundary entropy fluxes must then be included. For weak or renormalized solutions, an entropy inequality is established by approximation and lower semicontinuity; one cannot simply insert an unjustified logarithmic test into a weak equation.

Proposition 1.14 (Finite energy does not imply finite positive entropy)**.** *There is a probability density supported in $[0,1]^3$ with finite second moment and $\int f \log f = +\infty$.*

Proof. Let $c = 6/\pi^2$ and choose disjoint measurable sets A_n in $[0,1]^3$ with volumes $|A_n| = ce^{-n^2}/n^2$. Such sets exist because the sum of their volumes is less than one. Define $f = e^{n^2}$ on A_n and $f = 0$ elsewhere, with $0 \log 0 = 0$. Then

$$\int f = c\sum_{n\ge 1} n^{-2} = 1, \qquad \int |v|^2 f \, dv \le 3, \qquad \int f \log f = c\sum_{n\ge 1} 1 = +\infty.$$

□

This corrects a gap in the elementary entropy discussion in the reviewed Volkov slides [35]. Similarly, integrability of a continuous density alone does not force pointwise decay to zero at infinity: separated continuous spikes of height one and summable widths provide a counterexample. Decay assumptions used in integration by parts must therefore be stated, not inferred solely from normalization.

1.9.1 Entropy minimization is not a convergence proof

Proposition 1.15 (Maxwellian minimizes H at fixed first two moments)**.** *Let f be a probability density with mean u and $\int |v-u|^2 f \, dv = d\theta$, where $\theta > 0$. Set*

$$M_{u,\theta}(v) = (2\pi\theta)^{-d/2} e^{-|v-u|^2/(2\theta)}.$$

Whenever the entropy expressions are defined,

$$\int f \log f - \int M_{u,\theta} \log M_{u,\theta} = \int f \log \frac{f}{M_{u,\theta}} \ge 0.$$

Equality holds exactly when $f = M_{u,\theta}$ almost everywhere.

Proof. The function $\log M_{u,\theta}$ is a quadratic polynomial in v, so its integral against f agrees with its integral against $M_{u,\theta}$. This proves the identity. With $r = f/M_{u,\theta}$, use $r\log r - r + 1 \ge 0$ and integrate against $M_{u,\theta}\, dv$. Equality in this strictly convex inequality requires $r = 1$ almost everywhere. □

Under a positive kernel, sufficient continuity, and $\log f \in C^2$, vanishing $D(f)$ forces detailed balance $f'f'_* = ff_*$. Proposition 1.11 applied to $\log f$ then gives the Maxwellian form, with a negative quadratic coefficient required by integrability.

These facts characterize entropy minimizers and stationary collision equilibria. They do not prove that every trajectory converges to one. If a scalar function $H(t)$ is nonincreasing and bounded below, it has a limit, but its derivative need not tend to zero. For example, take a nonnegative smooth function b supported in $(-1/4, 1/4)$ with $b(0) = 1$ and define

$$h(t) = \sum_{n\geq 1} b(2^n(t-n)), \qquad H(t) = \int_t^\infty h(s)\,ds.$$

The bumps have disjoint supports and summable areas. Hence H is smooth, nonnegative, and decreasing to zero, while $H'(n) = -1$. Even $H'(t) \to 0$ alone would not supply compactness of the state or identify a unique limit. Convergence-to-equilibrium theorems require the appropriate coercivity, compactness, integrability, and control of invariants in the stated regime.

1.10 Exact moment balances and the closure problem

For a smooth unit-mass solution of

$$\partial_t f + v \cdot \nabla_x f = Q(f, f),$$

assume sufficient velocity integrability to justify the following moments. Let $c = v - u$ and define

$$P = \int c \otimes c\, f\, dv, \quad e = \frac{1}{2\rho}\int |c|^2 f\, dv, \quad q = \frac{1}{2}\int |c|^2 cf\, dv, \quad E = \frac{1}{2}\rho|u|^2 + \rho e.$$

Here P is a positive semidefinite pressure tensor. The Cauchy stress uses the opposite sign for the kinetic contribution; keeping this convention explicit avoids sign errors in viscosity formulas.

Proposition 1.16 (Moment equations before closure)**.** *Under these hypotheses,*

$$\partial_t \rho + \nabla \cdot (\rho u) = 0, \tag{1.14}$$

$$\partial_t(\rho u) + \nabla \cdot (\rho u \otimes u + P) = 0, \tag{1.15}$$

$$\partial_t E + \nabla \cdot (Eu + Pu + q) = 0. \tag{1.16}$$

Proof. Test the kinetic equation with $1, v, |v|^2/2$ and use the collision invariants. Since $\int cf\, dv = 0$, expansion of $v = u + c$ gives $\int v \otimes v f\, dv = \rho u \otimes u + P$. Likewise,

$$\frac{1}{2}\int |v|^2 v f\, dv = \left(\frac{1}{2}\rho|u|^2 + \rho e\right) u + Pu + q.$$

These expansions give all three equations. □

Example 1.17 (Equal five moments, different stress)**.** Let f_A and f_B be centered normalized Gaussian densities on $\mathbb{R}^3$ with covariance matrices

$$A = I, \qquad B = \operatorname{diag}(1+\delta, 1-\delta, 1), \quad 0 < \delta < 1.$$

They have the same mass, mean, and scalar temperature because $\operatorname{tr} A = \operatorname{tr} B = 3$. Their pressure tensors are A and B, respectively. Thus the five fluid moments do not determine the momentum flux.

A local Maxwellian imposes $P = \rho\theta I$ and $q = 0$, giving the Euler fluxes. An arbitrary local Maxwellian with space- and time-dependent parameters is generally not a solution of the full Boltzmann equation: its collision term vanishes, but its transport term need not. Euler equations arise as moment equations under that closure or as a justified limit, rather than as a proof that every Euler solution lifts to an exact local-Maxwellian kinetic solution. This qualification is stated explicitly in the rigorous hydrodynamic-limit literature [13].

1.10.1 First-order constitutive laws and scaling

For a monatomic gas near local equilibrium, a formal Chapman–Enskog expansion produces the constitutive form

$$P = pI - 2\mu S_0 - \zeta(\nabla \cdot u)I, \qquad q = -\kappa \nabla T, \quad S_0 = \tfrac{1}{2}(\nabla u + \nabla u^T) - \tfrac{1}{3}(\nabla \cdot u)I. \tag{1.17}$$

The bulk coefficient is zero in the simplest monatomic elastic model. Internal modes and other models change this conclusion. The values of the transport coefficients require solving equations for the linearized collision operator; conservation and symmetry alone do not fix them.

The Knudsen number is the ratio of a microscopic mean free path to the macroscopic length scale. Small Knudsen number, small Mach number, and fixed Reynolds number specify distinct relationships among scales. A formal expansion in one does not silently impose the others. A global small value also does not rule out local breakdown near a shock, boundary layer, or strongly varying rarefied region. Volkov's first and third decks make the physical scale separation and the failure of elementary closure in rarefied regimes explicit [33, 35].

1.10.2 A precise historical hydrodynamic-limit theorem

The result of Golse and Saint-Raymond that is useful here concerns the scaled equation on $\mathbb{R}^3_x \times \mathbb{R}^3_v$

$$\varepsilon^2 \partial_t F_\varepsilon + \varepsilon v \cdot \nabla_x F_\varepsilon = Q(F_\varepsilon, F_\varepsilon), \qquad M(v) = (2\pi)^{-3/2} e^{-|v|^2/2}. \tag{1.18}$$

Their collision kernels obey the exchange and reflection symmetries and hard-cutoff bounds of the form

$$0 < B(z,n) \le C(1+|z|)^\beta |\widehat{z} \cdot n|, \qquad \int_{S^2} B(z,n)\, dn \ge C^{-1} \frac{|z|}{1+|z|}, \quad 0 \le \beta \le 1,$$

with the inequalities understood almost everywhere. Initial data satisfy

$$\sup_{\varepsilon > 0} \varepsilon^{-2} H(F_\varepsilon^{\mathrm{in}} \mid M) < \infty, \quad H(F \mid M) = \iint \left(F \log \frac{F}{M} - F + M \right) dx\, dv. \tag{1.19}$$

Their Theorem 2.4 describes limit points of the fluctuations $g_\varepsilon = (F_\varepsilon/M - 1)/\varepsilon$ for the corresponding renormalized solutions. In the stated weak local topology the limit has the form

$$g(t,x,v) = u(t,x) \cdot v + \tfrac{1}{2}\theta(t,x)(|v|^2 - 5),$$

where (u, θ) solves an incompressible Navier–Stokes–Fourier system and satisfies its energy inequality [13, Theorem 2.4]. The initial velocity is determined by the weak limit of the Leray projection of the scaled momentum; a corresponding thermal moment determines the initial temperature fluctuation. For appropriate well-prepared initial entropy, the velocity is a Leray solution with the corresponding initial energy bound.

This is a theorem about a specified collision class, spatial domain, scaling, solution concept, and convergence topology. Its proof uses entropy bounds, coercivity of the linearized collision

operator, velocity averaging, vanishing conservation defects, and control of nonlinear fluxes. It yields weak fluid solutions; it does not by itself upgrade them to every stronger regularity or uniqueness statement. Statements about the separate 2026 proof release are analyzed in Chapter 2. That chapter's precise theorem must be used with its own forcing and solution hypotheses, rather than inferred from this historical limit theorem.

1.11 DSMC as a stochastic approximation, with auditable assumptions

The requested sixth Volkov deck explains direct simulation Monte Carlo as a particle approximation to kinetic dynamics, with statistical weights, free-flight and collision substeps, cell-based collision selection, inflow and wall kernels, and sampling of moments [38]. These simulated particles represent weighted samples. Their paths are not claimed to be the paths of every physical molecule in the gas.

Three error categories must be distinguished: the physical model (collision law and wall interaction), numerical discretization (time step, cells, particle count, cutoffs), and statistical estimation. Increasing a GPU sample count primarily addresses the latter two categories. It cannot validate a physical collision law by itself.

Proposition 1.18 (Flux sampling at an equilibrium wall)**.** *Let a stationary wall emit an equilibrium gas with thermal variance $\theta > 0$ and inward unit normal n. The normalized velocity density of molecules crossing the wall into the domain is*

$$p_{\mathrm{flux}}(v) = \frac{(v \cdot n)_+}{2\pi\theta^2} e^{-|v|^2/(2\theta)}.$$

Its positive normal component s has density $(s/\theta)e^{-s^2/(2\theta)}$, and its two tangential components are independent centered Gaussians of variance θ.

Proof. A bulk density $M_\theta(v)$ contributes a crossing flux $(v \cdot n)_+ M_\theta(v)$. The integral of this flux is $\sqrt{\theta/(2\pi)}$. Dividing by that value gives the displayed density. In orthonormal normal/tangential coordinates the exponential and its prefactor factor into the three stated one-dimensional densities. □

Sampling the normal velocity from an unweighted half-Gaussian would therefore give the wrong boundary ensemble. A finite reservoir implementation also requires an estimate of the missing high-speed tail; the Maxwellian has no finite maximum speed.

Proposition 1.19 (The exact frozen-rate majorant identity)**.** *Consider $K \geq 1$ possible events with fixed rates $r_j \geq 0$, and choose $R > 0$ with $R \geq \max_j r_j$. Generate a Poisson proposal process of rate KR, choose a label uniformly from $\{1, \ldots, K\}$ at each proposal, and accept label j with probability r_j/R. The accepted events of label j form a Poisson process of rate r_j; distinct accepted labels have independent counts.*

Proof. The probability that a proposal is accepted with label j is $r_j/(KR)$. For an interval of length t, condition on its Poisson number of proposals and take the joint probability generating function of the marked counts. Summing the Poisson series gives

$$\exp\left(t \sum_{j=1}^{K} r_j(z_j - 1)\right) = \prod_{j=1}^{K} \exp(t r_j(z_j - 1)).$$

This is the product of the generating functions of independent Poisson variables with the claimed means. The same argument on disjoint time intervals proves the process statement. □

This proves the sampling principle behind majorant collision schemes. It does not alone prove convergence of a full DSMC method whose rates change after collisions and whose transport is split in time. In that case the bound must remain valid for the current state and the discretization error must be controlled. If an alleged acceptance probability exceeds one, silently clipping it changes the rate. A temperature-based constant in illustrative code is not a proof of a global bound on an unbounded velocity distribution.

Proposition 1.20 (Sampling variance with correlations)**.** *If $Y_1, \dots, Y_n$ have finite second moments, then*

$$\operatorname{Var}\left(\frac{1}{n}\sum_i Y_i\right) = \frac{1}{n^2}\sum_{i,j}\operatorname{Cov}(Y_i, Y_j).$$

For identically distributed independent samples this equals σ^2/n. For a stationary sequence with variance $\sigma^2 > 0$ and lag correlations r_k, it equals

$$\frac{\sigma^2}{n}\left(1 + 2\sum_{k=1}^{n-1}(1 - k/n)r_k\right).$$

Proof. Expand the square of the centered sum and take expectations. For a stationary sequence, group off-diagonal pairs by their lag; there are $n - k$ pairs at lag k in each direction. □

Thinning a time series can reduce some correlations but does not prove independence. A central-limit error bar requires its own assumptions; a fixed seed provides reproducibility, not randomness certification. For independent samples, Chebyshev's inequality gives the explicit bound $\Pr(|\overline{Y} - \mathbb{E}Y| \geq a) \leq \sigma^2/(na^2)$. The statistics and Monte Carlo decks are useful introductions [36, 37]; implementations should additionally handle finite precision, zero arguments of logarithms, array bounds, correct random-number-generator state, and reproducible restart state. The period of a generator is constrained by its internal state space, not merely by the number of possible values of a single output word.

1.12 Reduced models and learned closures

An orthogonal reduced basis minimizes a chosen projection loss on its training snapshots. That statement does not imply that the projected trajectory is stable, that the discarded modes exert no force, or that the basis will predict new parameter regimes. The currently available replacement Farhat course decks explicitly distinguish projection error from model-dynamics error [10]. Their complete version-specific review is recorded separately in the source audit.

For an orthonormal column matrix V, write a full state $z = Va + w$ with $V^T w = 0$. If $\dot{z} = F(z)$, the exact retained coordinate satisfies $\dot{a} = V^T F(Va + w)$. Replacing the right side by $V^T F(Va)$ is a closure. The exact memory identity in Chapter 4 explains why eliminated coordinates can leave both memory and an initial-condition contribution.

Proposition 1.21 (What weighted POD actually minimizes)**.** *Let $X = [x_1 \ \cdots \ x_s] \in \mathbb{R}^{n\times s}$, let $W = \operatorname{diag}(w_1, \dots, w_s)$ with $w_j > 0$, and let $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$ be the eigenvalues of $C = XWX^T$. Fix $1 \leq r \leq n$. Among all $V \in \mathbb{R}^{n\times r}$ with $V^T V = I$,*

$$\min_V \sum_j w_j \|x_j - VV^T x_j\|^2 = \sum_{i>r} \lambda_i.$$

A minimizing range is spanned by eigenvectors for the largest r eigenvalues.

Proof. The objective is $\operatorname{tr} C - \operatorname{tr}(V^T C V)$. In an orthonormal eigenbasis e_i of C, put $a_i = \|V^T e_i\|^2$. Then $0 \le a_i \le 1$, $\sum_i a_i = r$, and $\operatorname{tr}(V^T C V) = \sum_i \lambda_i a_i$. Transferring any weight from a smaller eigenvalue to an unfilled larger one cannot decrease this sum. Its maximum is therefore $\sum_{i \le r} \lambda_i$, and the indicated eigenspace attains it. □

The proposition concerns the chosen snapshots and weights. It contains no assertion about an unsampled parameter or a differential equation. Indeed the stable matrix

$$A = \begin{pmatrix} -1 & 10 \\ 0 & -1 \end{pmatrix}, \qquad V = 2^{-1/2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

has both eigenvalues equal to -1, while its orthogonal Galerkin reduction is $V^T A V = 4$. The reduced scalar solution grows exponentially. Nonsymmetric stable dynamics are therefore not automatically preserved by orthogonal projection.

For Petrov–Galerkin projection with full-column-rank trial and test matrices V, W, the matrix $W^T V$ must be invertible to use $\Pi = V(W^T V)^{-1} W^T$. Direct multiplication gives $\Pi^2 = \Pi$, $\operatorname{range} \Pi = \operatorname{range} V$, and $\ker \Pi = \ker W^T = (\operatorname{range} W)^\perp$. The direction of this oblique projection is the latter kernel.

1.12.1 Hyperreduction adds another controlled approximation

State reduction need not make a nonlinear residual cheap to evaluate. Let $U \in \mathbb{R}^{n \times r}$ have orthonormal columns and let $P \in \mathbb{R}^{n \times r}$ select r coordinates. If $P^T U$ is invertible, the discrete empirical interpolation map is

$$D = U(P^T U)^{-1} P^T.$$

Proposition 1.22 (Interpolation exactness and amplification)**.** *For every $y \in \mathbb{R}^n$, $DU = U$, $P^T D y = P^T y$, and*

$$\|y - Dy\| \le \left(1 + \|(P^T U)^{-1}\|\right) \|(I - UU^T) y\|.$$

Proof. The first two identities follow by cancellation of $P^T U$. Because D is the identity on the range of U, $(I - D)y = (I - D)(I - UU^T)y$. Use $\|I - D\| \le 1 + \|D\|$ and the unit operator norms of U and P^T to obtain the estimate. □

The bound is deliberately elementary; sharper oblique-projection bounds are possible. Even this version shows why a small discarded snapshot norm is insufficient if the sampled basis is ill-conditioned. For oversampled least squares, full column rank of $P^T U$ is the corresponding uniqueness and exact-reproduction requirement. An online cost independent of n additionally requires that sampled nonlinear entries and the restricted reconstruction can be evaluated without assembling all n entries. A dense nonlocal nonlinear map can violate that requirement.

Other reduction methods have different hypotheses. Classical balanced truncation concerns stable finite-dimensional linear time-invariant systems, with reachability and observability conditions, and its usual input–output error estimate assumes zero initial state. An empirical balanced-POD approximation does not inherit every exact balancing guarantee. Moment matching requires the relevant full and reduced resolvents to exist at its interpolation nodes. Grassmann-chart interpolation requires the chart to be valid, fixed subspace rank, and appropriate alignment before interpolating reduced operators. The detailed Farhat review documents these distinctions [10].

Likewise, a weighted finite-element energy $\widetilde{U}(q) = \sum_e \xi_e U_e(q)$ with fixed weights has gradient $\sum_e \xi_e \nabla U_e(q)$. If a neural model makes $\xi_e = \xi_e(q)$, the gradient instead includes $\sum_e U_e(q) \nabla \xi_e(q)$.

Omitting those terms loses the claimed potential-force relation. A preservation theorem for a fixed-weight mechanical hyperreduction must therefore be re-proved when the weights become state dependent or the equation changes to a different fluid formulation.

1.12.2 A provable constraint for a neural constitutive model

A classical or quantum neural model can be used to predict constitutive coefficients from invariant local features. For example, a model can output nonnegative scalars μ, ζ, κ in (1.17). This architecture has a limited but useful exact guarantee.

Proposition 1.23 (Nonnegative constitutive entropy production)**.** *For $T > 0$, let the viscous stress be $\tau = 2\mu S_0 + \zeta(\nabla \cdot u)I$ and the heat flux $q = -\kappa \nabla T$, with $\mu, \zeta, \kappa \geq 0$. Then*

$$\frac{\tau : \nabla u}{T} - \frac{q \cdot \nabla T}{T^2} = \frac{2\mu |S_0|^2}{T} + \frac{\zeta(\nabla \cdot u)^2}{T} + \frac{\kappa |\nabla T|^2}{T^2} \geq 0.$$

Proof. The symmetric stress is orthogonal in the Frobenius inner product to the antisymmetric part of ∇u. The traceless symmetric part S_0 is orthogonal to I. Decompose the symmetric part of ∇u into $S_0 + (\nabla \cdot u)I/3$ and substitute the heat-flux formula. □

If the scalar inputs are rotational invariants, these constitutive tensors transform correctly under rotations. Dependence on gradients and thermodynamic invariants also permits Galilean covariance. Neither property proves accurate transport coefficients. Moreover, nonnegative entropy production does not by itself make $P = pI - \tau$ positive semidefinite far from equilibrium. A kinetic moment model that requires realizable covariances can instead parameterize $P = LL^T$ or another positive-definite form and impose the remaining energy and flux constraints explicitly. These are different requirements and must be checked together.

Tensor decomposition makes these constraints transparent. In three dimensions a rank-two tensor splits into scalar trace, antisymmetric, and symmetric traceless pieces, of dimensions $1, 3, 5$. The antisymmetric piece is an exterior-square representation; the symmetric pieces arise from symmetric tensor powers and their trace decomposition. Schur functors and invariant contractions therefore help build features and outputs with known transformation laws. Determinants test nonsingularity or volume change in specified finite-dimensional maps. They do not determine a turbulent closure or imply the analytic estimates required for a continuum theorem.

For a quantum version of ToricGT, the surface or dessin specifies connectivity and admissible local operations, while circuit parameters specify a family of quantum channels and observables. A conservation law can be enforced by restricting channels or observables to a specified invariant sector. A measured expectation then estimates a well-defined feature or coefficient. State preparation, finite-shot variance, circuit approximation, and the relation between that feature and the physical kinetic state remain separate parts of the model. The quantum-fluid chapter discusses these algorithmic costs and encodings. Learning on a surface does not identify an arbitrary three-dimensional fluid with a two-dimensional toric code.

1.13 What passes from complexes to a continuum theorem

On a finite cell complex, the boundary-of-boundary identity supplies exact incidence relations. Its adjoint cochain identities, Hodge decompositions, and positive semidefinite Laplacians can enforce discrete conservation and isolate harmonic components. A gentle surface algebra or a dessin provides additional combinatorial organization of local operators. A variety of complexes organizes matrices

with prescribed compositions and rank conditions. These constructions can certify the algebraic structure of a discretization or constrain a neural architecture.

To pass from such a model to a continuum claim requires a separate chain of estimates: consistency of the geometric approximation, stability uniform in mesh size, compactness in a topology suitable for nonlinear products, identification of the limiting equation, and preservation of the required initial and boundary data. Spectral gaps on individual finite graphs need not have a uniform positive lower bound as the graphs are refined. Rank constraints on finitely many matrices need not control high-frequency concentration. A learned closure with positive entropy production need not supply the required continuum regularity.

The constructive role of the project's algebra is to make some requirements exact and testable, then to identify which analytic requirements are still independent. Chapter 3 proves the finite structural statements, and Chapter 2 identifies the actual theorem and proof obligations of the recent release. The microscopic derivations in this chapter provide the physical interpretation and the correct hierarchy of approximations without changing those theorems' hypotheses.

Chapter 2

The Navier–Stokes release: exact scope and proof interfaces

Evidence status. The 166-page released manuscript has been read in full across the two mathematical review passes recorded in the source ledger. The repository revision, final theorem statements, dependency graph, and selected implementation modules have been inspected. Reading a mathematical proof, rebuilding a Lean target, and independently checking an exported kernel certificate are distinct operations; their actual outcomes are recorded separately. The propositions proved in this chapter explain mathematical interfaces of the construction. They do not replace the analytic construction imported from the release.

2.1 What the released theorem says

OpenAI's release of 8 September 2026 concerns a smooth, externally forced, three-dimensional incompressible flow. It supplies a manuscript and Lean formalization. The Clay Mathematics Institute's response of 11 September acknowledges the apparent resolution while describing an ongoing evaluation process. These facts must be distinguished from an award decision [7, 27].

Let $\nu > 0$ and define the momentum residual

$$\mathcal{R}_\nu(u,p) = \partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p. \tag{2.1}$$

The source's main conclusion can be recorded without changing its quantifiers:

$$\begin{aligned} &f \in C_c^\infty(\mathbb{R}^3\times(0,\infty);\mathbb{R}^3), \qquad u,p \in C^\infty(\mathbb{R}^3\times[0,1)),\\ &\mathcal{R}_\nu(u,p) = f, \quad \nabla\cdot u = 0, \quad u(\cdot,0)=0, \qquad (2.2)\\ &\operatorname{supp} u(\cdot,t)\cup\operatorname{supp} p(\cdot,t)\subset K \quad (0\le t<1),\\ &\sup_{t<1}\|u(t)\|_2<\infty, \qquad \limsup_{t\uparrow 1}\|u(t)\|_\infty=\infty. \qquad (2.3)\end{aligned}$$

Here K is compact. With the same datum and force there is no global smooth whole-space solution with uniformly bounded kinetic energy. A corresponding periodic result is also established [26, Theorem 1.1 and Corollary 10.6].

In Fefferman's formulation, alternatives (A) and (B) impose $f = 0$, whereas (C) and (D) permit a smooth force satisfying the specified decay conditions. The released statements address (C) and (D). They do not logically negate (A) or (B): an existential assertion with $f \neq 0$ does not supply an example with $f = 0$. Pressure periodicity is part of the periodic formulation, including its erratum

[11]. The separate Euler result concerns a different equation, obtained by setting viscosity to zero, and cannot be transferred to positive viscosity by deleting the viscous estimates.

This distinction changes the appropriate goal of the present extension. A more informative presentation can expose the cancellation, support, analytic norms, and formal dependency interfaces. A stronger theorem would require a new mathematical conclusion with all its additional hypotheses discharged. The material below provides the former and identifies precisely where the source's construction supplies the latter's indispensable analytic inputs.

2.2 An exact algebra for residual correction

The same distinction between a complex and an exact complex encountered earlier reappears here. A differential identity constrains what a correction can do. It does not prove that a correction of the necessary size, support, and regularity exists.

Proposition 2.1 (Quadratic residual identity)**.** *For smooth u, w, p, π, with $\nabla \cdot w = 0$,*

$$\mathcal{R}_\nu(u + w, p + \pi) = \mathcal{R}_\nu(u, p) + L_{u,\nu}(w, \pi) + \nabla \cdot (w \otimes w),$$
$$L_{u,\nu}(w, \pi) = \partial_t w + (u \cdot \nabla)w + (w \cdot \nabla)u - \nu\Delta w + \nabla\pi.$$

The tensor convention is $(w \otimes w)_{ij} = w_i w_j$ and $(\nabla \cdot S)_i = \sum_j \partial_j S_{ij}$.

Proof. All terms other than advection are linear. Expanding the latter gives $(u + w)_j \partial_j (u_i + w_i)$, with the four products obtained by choosing one of u, w in each factor. Moreover, $\sum_j \partial_j (w_i w_j) = \sum_j w_j \partial_j w_i + w_i \sum_j \partial_j w_j = (w \cdot \nabla)w_i$. This proves every component of the identity. No averaging, limiting argument, or neglected term occurs. □

In particular, solving $L_{u,\nu}(w, \pi) = -\mathcal{R}_\nu(u, p)$ leaves the quadratic residual $\nabla \cdot (w \otimes w)$. A numerical decrease in the linear residual alone is therefore an incomplete verification. Conversely, if a background satisfies $\mathcal{R}_\nu(u, p) = -\nabla \cdot S + E$, a fluctuation with averaged covariance S can cancel the selected mean stress while creating other linear and oscillatory terms. Each of those terms needs a place in the next residual estimate.

Proposition 2.2 (A transverse wave and its covariance)**.** *Let $a, k \in \mathbb{R}^3$ with $a \cdot k = 0$ and put $w(x, \phi) = a \cos(k \cdot x + \phi)$. Then*

$$\nabla \cdot w = 0, \qquad (w \cdot \nabla)w = 0, \qquad \frac{1}{2\pi} \int_0^{2\pi} w \otimes w \, d\phi = \frac{1}{2} a \otimes a.$$

The vanishing self-advection statement requires constant a and k.

Proof. Differentiating the cosine gives $\nabla \cdot w = -(a \cdot k) \sin(k \cdot x + \phi)$ and $(w \cdot \nabla)w = -(a \cdot k)a \cos(k \cdot x + \phi) \sin(k \cdot x + \phi)$. Both vanish. Integration of $\cos^2 s = (1 + \cos 2s)/2$ proves the tensor identity. If the amplitude or phase gradient varies, the product rule introduces additional derivatives, so this proof gives no cancellation for them. □

The covariance is an element of $\mathrm{Sym}^2(\mathbb{R}^3)$, with nonnegative quadratic form $v \mapsto (a \cdot v)^2/2$. The positivity is a concrete use of a symmetric tensor power. A rank-one tensor representation does not by itself solve the vector-valued stress-divergence equation; the allowed polarization directions and the differential constraints still matter. The exterior-power test for independence of two stress directions is developed in Chapter 3.

Proposition 2.3 (Averaging and nonlinear closure)**.** *Let P be a linear averaging operator on a function algebra with $P1 = 1$. Assume $P^2 = P$ and the conditional-expectation identity $P(ag) = aP(g)$ whenever $Pa = a$. If $v = Pu$ and $w = u - v$, then*

$$P(u \otimes u) = v \otimes v + P(w \otimes w).$$

If P is positive, the final term is positive semidefinite pointwise in the averaged variables.

Proof. Idempotence gives $Pw = 0$. Expand $(v + w) \otimes (v + w)$. The two mixed terms have zero average by the assumed multiplication identity, and $P(v_i v_j) = v_i v_j$. For a constant test vector b, $b^T P(w \otimes w)b = P((b \cdot w)^2) \geq 0$. □

An orthogonal projection onto an arbitrary collection of modes need not have the multiplication property in this proposition. Likewise, averaging an extended function $F(x, Y)$ over independent Y and then substituting a phase $Y = \Phi(x)$ is different from first restricting and averaging the resulting physical function. If $F(x, Y) = \cos(2\pi(Y - x))$ on one-dimensional tori, then $\int F(x, Y)dY = 0$, whereas $F(x, x) = 1$. This elementary example prevents an unjustified replacement of a pointwise physical residual by an auxiliary mean. The phase derivatives and support geometry in a complete construction are precisely what allow these two kinds of expression to be related.

2.3 Concentration, energy, and the information lost by norms

Proposition 2.4 (Energy estimate from rest)**.** *Let u, p be a smooth solution on $[0, T]$ of $\mathcal{R}_\nu(u, p) = f$, $\nabla \cdot u = 0$, $u(0) = 0$. Assume periodicity, or sufficient decay for the integrations by parts below; fixed compact spatial support of u, p is sufficient. If $F(t) = \int_0^t \|f(s)\|_2 \, ds$, then*

$$\|u(t)\|_2^2 + 2\nu \int_0^t \|\nabla u(s)\|_2^2 \, ds \leq F(t)^2.$$

Proof. The pressure pairing is $\int u \cdot \nabla p = -\int p \nabla \cdot u = 0$. Advection contributes $\int u \cdot (u \cdot \nabla)u = \frac{1}{2} \int \nabla \cdot (u|u|^2) = 0$. Thus

$$\frac{1}{2} \frac{d}{dt} \|u\|_2^2 + \nu \|\nabla u\|_2^2 = \langle f, u \rangle.$$

Write $y_\delta = (\|u\|_2^2 + \delta^2)^{1/2}$, $\delta > 0$. Then $y'_\delta \leq \|f\|_2$, so $y_\delta(t) \leq \delta + F(t)$. Passing to $\delta \downarrow 0$ gives $\|u(t)\|_2 \leq F(t)$. Integrating the energy identity once more,

$$\|u(t)\|_2^2 + 2\nu \int_0^t \|\nabla u\|_2^2 \leq 2 \int_0^t F(s)F'(s) \, ds = F(t)^2.$$

The argument on each $T < 1$, followed by monotone convergence of the nonnegative dissipation integral, also applies up to a singular time. □

This estimate controls an integral, not the maximum of the integrand. Information reduction becomes mathematically consequential at this exact point: a small spatial set can carry a large amplitude while making a small contribution to L^2 energy.

Proposition 2.5 (Anisotropic concentration test)**.** *Let $0 < h < 1/6$, $\tau > 0$, and $W \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^m)$ be nonzero. Define the comparison family*

$$V_\tau(x_1, x_2, x_3) = \tau^{-1/2-h} W\left(\frac{x_1}{\tau^{1/2}}, \frac{x_2}{\tau^{1/2}}, \frac{x_3}{\tau^{1/2-h}}\right).$$

For $1 \le p < \infty$,

$$\|V_\tau\|_p^p = \tau^{3/2-h-p(1/2+h)}\|W\|_p^p, \qquad \|V_\tau\|_\infty = \tau^{-1/2-h}\|W\|_\infty.$$

Its energy tends to zero, while its maximal amplitude diverges. Moreover, $\int_0^{\tau_0} \|\nabla V_\tau\|_2^2 \, d\tau < \infty$ for every finite $\tau_0 > 0$. This comparison family is not asserted to solve Navier–Stokes.

Proof. The change of variables has determinant $\tau^{3/2-h}$. This proves the norm identities. At $p = 2$ the exponent is $1/2 - 3h > 0$. Each of the first two spatial derivatives contributes $\tau^{-1/2}$, so its squared norm has exponent $-1/2 - 3h > -1$. The axial derivative contributes $\tau^{-1/2+h}$ and has squared-norm exponent $-1/2 - h > -1$. Both powers are integrable at zero. The supremum formula follows because the coordinate dilation is onto. □

For this family the critical spatial integrability exponent is

$$p_*(h) = \frac{3/2 - h}{1/2 + h} < 3.$$

At $p = 3$, $\|V_\tau\|_3^3 = \tau^{-4h}\|W\|_3^3$, which diverges. Thus the scaling test does not contradict a regularity theorem requiring a bounded critical L^3 norm. Such a theorem is a substantially stronger statement than an energy bound; the endpoint theorem for the unforced Cauchy problem is due to Escauriaza, Seregin, and Šverák [9]. Our comparison family proves only the logical insufficiency of the weaker norm information, independently of any existence construction.

2.4 Similarity coordinates as an exact change of variables

The geometry relevant to the release can be organized using the positive variables

$$\tau = 1 - t, \qquad D = \tfrac{1}{2} - h, \qquad \tau = q(1 - \eta^2), \quad z = q^D \eta, \quad X = \frac{r^2}{2q}, \quad 0 < h < \tfrac{1}{2}.$$

The next calculation verifies the differential interface rather than assuming that symbolic substitution preserves all derivatives.

Proposition 2.6 (Coordinate existence and derivative weights)**.** *For $\tau > 0$ and $z \in \mathbb{R}$ there is a unique $q > |z|^{1/D}$ with $q - z^2 q^{2h} = \tau$. Set $\eta = zq^{-D}$, $d = 1 - \eta^2$, and $L = 1 - 2h\eta^2$. Then $|\eta| < 1$, $L \ge 1 - 2h > 0$, and for a smooth profile $G(X, \eta)$ and any real b,*

$$\partial_t(q^b G) = q^{b-1} L^{-1}(-bG + D\eta\partial_\eta G + X\partial_X G),$$
$$\partial_z(q^b G) = q^{b-D} L^{-1}(2b\eta G + d\partial_\eta G - 2\eta X\partial_X G).$$

Proof. On $q > |z|^{1/D}$, the function $q - z^2 q^{2h}$ begins at zero, tends to infinity, and has derivative $1 - 2hz^2 q^{2h-1} \ge 1 - 2h > 0$. The intermediate-value theorem and strict monotonicity give existence and uniqueness. The implicit-function theorem gives smoothness, and differentiation yields

$$q_t = -L^{-1}, \quad q_z = 2\eta q^{1-D}/L, \quad \eta_t = D\eta/(qL), \quad \eta_z = d/(q^D L),$$

$$X_t = X/(qL), \qquad X_z = -2\eta X/(q^D L).$$

The two formulas follow by the product and chain rules. The endpoint $z = 0$ also satisfies them, because the implicit derivative remains nonzero. □

On bounded profile ranges, a transverse derivative has weight $q^{-1/2}$, an axial derivative has weight q^{-D}, and a time derivative has weight q^{-1}. The denominators remain bounded because $L \geq 1 - 2h$. Constants can depend on the derivative order and fixed profile parameters. They cannot be allowed to hide an extra loss growing with the approximation stage if one later intends to sum an infinite correction sequence.

The apparent cylindrical singularity at $r = 0$ must also be resolved. For instance, if the azimuthal velocity is $r\, a(r^2, z, t)$, then

$$r\, a(r^2, z, t) e_\theta = a(x_1^2 + x_2^2, z, t)(-x_2, x_1, 0)$$

is smooth in Cartesian coordinates. Boundedness of a cylindrical component alone would not establish this. This is a local regularity condition on a representation, comparable to requiring that coordinate expressions of a section glue smoothly; it is not supplied by the topology of the domain.

2.5 A quadratic relation that controls analytic derivative loss

An especially useful connection to the project's algebraic methods occurs in the analytic linear recursion. The relevant feature is a forbidden composition of derivative-carrying blocks. We isolate its mechanism in a statement independent of fluid terminology.

Lemma 2.7 (Words with no adjacent derivative letters)**.** *A word of length k in the letters N, D with no consecutive D has at most $\lceil k/2 \rceil$ occurrences of D. The number with exactly j occurrences is $\binom{k-j+1}{j}$, interpreted as zero when $j > k - j + 1$. The total number a_k satisfies $a_0 = 1$, $a_1 = 2$, and $a_k = a_{k-1} + a_{k-2}$ for $k \geq 2$.*

Proof. Between j derivative letters there are at least $j - 1$ other letters, so $2j - 1 \leq k$. If their positions are $1 \leq i_1 < \cdots < i_j \leq k$ with $i_{r+1} \geq i_r + 2$, the shifted positions $i_r - (r - 1)$ form an ordinary j-element subset of $\{1, \ldots, k - j + 1\}$. This is a bijection, proving the binomial formula. For the recurrence, an admissible word ending in N is an arbitrary admissible word of length $k - 1$ followed by N. One ending in D, when $k \geq 2$, ends in ND and corresponds to a word of length $k - 2$. These two classes are disjoint and exhaustive. □

Proposition 2.8 (Sparse Volterra systems)**.** *Let $E = E_0 \oplus E_1$ be a finite-dimensional normed complex vector space. Fix a compact real interval I and let $S_\rho = \{z \in \mathbb{C} : \operatorname{dist}(z, I) < \rho\}$. Consider an integral equation on $0 \leq s \leq a$ and this complex parameter neighborhood,*

$$W = Gf + G(A_0 W + A_1 \partial_\eta W), \qquad (Gg)(s) = \int_0^s K(s, r) g(r)\, dr.$$

Assume $\|K(s, r)\| \leq 1$, K preserves the two summands and is independent of η; the coefficients and kernel are continuous in their radial variables; A_0, A_1, f are uniformly bounded and holomorphic in η; and every value and every η-derivative of A_1 maps E_0 into E_1 and annihilates E_1. On each smaller neighborhood $S_{\rho'}$ with $0 < \rho' < \rho$, the Picard series converges for every finite a. It gives the unique solution in the class of solutions bounded and holomorphic on a parameter neighborhood strictly larger than the one on which they are compared.

Proof. Expand the k-fold iterate of $G(A_0 + A_1 \partial_\eta)$ into words. The kernel between adjacent coefficients preserves E_0, E_1. Therefore

$$A_1(s) K(s, r) A_1(r) = 0, \qquad A_1(s) K(s, r) \partial_\eta A_1(r) = 0.$$

The second identity is necessary: in a composition of two derivative operators the first derivative can strike the second coefficient. It vanishes for the same block reason. Thus a nonzero word cannot contain two adjacent derivative operators, even after expanding the product rule. By Lemma 2.7, at most $p_k = \lceil k/2 \rceil$ Cauchy radius losses are required.

Put $\Delta = \rho - \rho' > 0$. Distribute that loss in p_k equal parts. The Cauchy estimate for one parameter derivative costs at most p_k/Δ; when $p_k = 0$ use cost one. Operator norms absorb the coefficient bounds, and the nested integration domain has volume $a^{k+1}/(k+1)!$. Bounding the number of words by 2^k gives, after enlarging a constant C,

$$\left\| [G(A_0 + A_1\partial_\eta)]^k Gf \right\|_{S_{\rho'}} \le \frac{C^{k+1} a^{k+1}}{(k+1)!} \max\{1, p_k/\Delta\}^{p_k}.$$

Stirling's bound shows that the kth root of the right-hand side is $O(k^{-1/2})$, tending to zero. Hence the series converges uniformly on the smaller neighborhood. Applying the same estimate after one additional radius restriction permits differentiation in η and termwise substitution into the integral equation. If $W_1 - W_2 = G(A_0 + A_1\partial_\eta)(W_1 - W_2)$, iteration and the same estimate make the difference zero on the smaller neighborhood. Analytic continuation gives the stated uniqueness. □

The source's six-component linear recursion has the splitting $E_0 = \mathbb{C}^4$, $E_1 = \mathbb{C}^2$, with diagonal radial kernels. Its use of the above relation is traceable to Lemma 5.1, rather than inferred from a superficial resemblance to a quiver [26]. The relation has three precise interpretations. It eliminates forbidden operator words; it limits derivative loss; and, with the integration simplex, it makes a convergent analytic solve possible on a common radial interval.

The language of admissible words is regular: a finite automaton needs only to remember whether the preceding letter was D, plus a rejecting state. The analytic conclusion nevertheless requires unbounded iteration and quantitative estimates. A regular syntax does not make the represented analytic task finite-dimensional.

Nor does this calculation establish a gentle algebra. The abstract one-vertex algebra $k\langle N, D\rangle/(D^2)$ is infinite-dimensional, and the arrow N has two allowed successors. It therefore fails a usual gentle-algebra continuation condition. The honest bridge is to a representation with a quadratic zero relation and a controlled path language. Any claim of a particular gentle presentation would require a separate equivalence or construction.

2.6 From formal orders to actual smooth fields

The order of a term in a formal expansion must not be confused with a convergence estimate for the coefficients. Suppose coefficients of order j have constants growing faster than every exponential. Positive powers of a small parameter do not automatically make their unrestricted sum converge. A suitable realization uses cutoffs chosen after the coefficients are known.

Lemma 2.9 (A diagonal realization estimate)**.** *Let $0 < q < q_0 \le 1$ on an open domain, with q smooth and bounded below on every compact subset. Suppose smooth increments Z_j, $j \ge 1$, satisfy*

$$|Z_j|_m \le C_{j,m} q^{g_j - \ell_m} (1 + |\log q|)^{P_{j,m}}, \qquad 0 < g_1 \le g_2 \le \cdots \longrightarrow \infty,$$

where $|\cdot|_m$ is the pointwise maximum of coordinate derivatives through order m, and ℓ_m is independent of j. Assume that differentiating a cutoff $\chi(aq)$ costs a power of q^{-1} depending only on derivative

order, uniformly in a on its transition region. Then one can choose $a_{j+1} \geq 2a_j$ so that the cutoff sum $Z = \sum_{j\geq 1} \chi(a_j q) Z_j$ is locally finite and smooth, and its tail after a fixed initial block satisfies

$$\left| Z - \sum_{j=1}^{J} Z_j \right|_m \leq 2^{-J} q^{g_{J+1}/2 - L_m} \quad \left(J \geq \max\{1, m\},\ q < \frac{1}{2a_J} \right).$$

Here $\chi = 1$ on $[0, 1/2]$, $\chi = 0$ on $[1, \infty)$, and L_m is independent of J.

Proof. The product rule and the cutoff hypothesis bound the mth seminorm of the jth summand by $\widehat{C}_{j,m} q^{g_j - L_m} (1 + |\log q|)^{\widehat{P}_{j,m}}$. At stage j impose this finite list of requirements for $0 \leq m \leq j$:

$$\widehat{C}_{j,m} q^{g_j/2} (1 + |\log q|)^{\widehat{P}_{j,m}} \leq 2^{-j} \quad (0 < q \leq a_j^{-1}).$$

It is possible because every positive power of q dominates every fixed power of $|\log q|$ at zero. Increase a_j further to preserve its geometric growth. On a compact subset where $q \geq \delta > 0$, terms with $a_j > \delta^{-1}$ vanish, proving local finiteness and smoothness. If $q < (2a_J)^{-1}$, the first J cutoffs equal one. For $j > J \geq m$, the bound just imposed and $g_j \geq g_{J+1}$ give

$$\sum_{j>J} 2^{-j} q^{g_j/2 - L_m} \leq 2^{-J} q^{g_{J+1}/2 - L_m},$$

as required. □

To preserve a differential constraint, apply the cutoffs to suitable potentials. In three-dimensional Euclidean space,

$$\nabla \times (\chi(a_j q) A_j) = \chi(a_j q) \nabla \times A_j + a_j \chi'(a_j q) \nabla q \times A_j.$$

The second term is a real part of the field and must be retained. Taking divergence of the full curl yields zero. Cutting off an already divergence-free velocity directly would instead give $\nabla \cdot (\chi u) = \nabla \chi \cdot u$, which need not vanish.

Proposition 2.10 (All-order residual realization)**.** *Let $U^{[J]}$ be finite approximations and U their cutoff realization. Assume their derivatives have polynomial growth in q^{-1} with exponent losses independent of J, and that the tail estimate of Lemma 2.9 applies through every fixed order. Let F be a fixed differential polynomial of finite order and degree whose coefficient derivatives have polynomial growth in q^{-1}. If*

$$|F(U^{[J]})|_m \leq C_{J,m} q^{\rho_J - K_m} (1 + |\log q|)^{P_{J,m}} + E_{J,m}, \quad \rho_J \to \infty,$$

with K_m independent of J and each fixed $E_{J,m} = O(q^N)$ for every N, then $|F(U)|_m = O(q^N)$ for every m, N.

Proof. Fix m, N. Let s, d bound the differential order and polynomial degree of F. For products use the exact identity

$$\prod_{i=1}^{d} a_i - \prod_{i=1}^{d} b_i = \sum_{r=1}^{d} \left(\prod_{i<r} a_i \right) (a_r - b_r) \left(\prod_{i>r} b_i \right).$$

The product rule then bounds $F(U) - F(U^{[J]})$ in seminorm m by a fixed power of q^{-1} times the tail seminorm $m + s$ and a polynomial in the two field seminorms $m + s$. All exponent losses depend on m, s, d, not J. Choose J large enough that both ρ_J and $g_{J+1}/2$ exceed N plus these losses and one extra unit. Keep that J fixed. Shrink the q-neighborhood so all its finitely many constants and logarithms are absorbed by that extra unit, and the first J cutoffs equal one. The finite residual, its fixed flat remainder, and the residual difference are then all $O(q^N)$. □

The order of these quantifiers is essential:

$$\forall(m,N)\ \exists J\ \exists\delta,C\ \forall q\in(0,\delta):\ |F(U)|_m\le Cq^N.$$

It does not assert one J that controls every derivative. It does not require summing the flat remainders $E_{J,m}$ over J. It also does not assert convergence of the uncut formal series. This distinction is the analytic counterpart of keeping a filtered algebra, its associated graded object, and a completed realization separate.

2.7 The finite correction cycle as a verifiable dependency graph

The source's construction provides the hypotheses needed above through separate analytic modules. The following locator table records where to check them; the table is a dependency guide, not a substitute proof [25, 26].

Interface	Mathematical obligation	Source location
Background	Smooth axis; exact exterior; supported compatible stress; positive cone margins	Theorem 4.6; Appendices A–C
Order recursion	A common radial interval, moment repair, derivative losses independent of formal order	Lemmas 5.1–5.4; Proposition 5.5
Phase geometry	Support separation and compatible finite covering coordinates	Section 6
Oscillatory solve	Transverse evolution, damping, real covariance, full curl correction	Propositions 7.2, 7.5–7.6; Lemma 7.7
Mean solve	Compact primitives, auxiliary inverse, five moment equations	Section 8
Iteration	Full residual recomputation, fixed-domain gains, all physical derivatives	Propositions 9.3–9.9
Completion	Smooth forcing, energy, comparison, all viscosities, periodization	Section 10

One can encode a correction state as a typed record rather than a loose list of estimated errors. Its fields should include a velocity, pressure, potential representation, support certificates, exact preserved moments, residual decomposition, and derivative bounds. A correction operation must produce a new such record. The pressure and residual are recomputed from the updated velocity, so one cannot reuse a previous covariance after altering a wave amplitude. This record-based reading is directly compatible with the typed graph approach developed earlier in the monograph.

The numerical exponents also permit an elementary exact check. Put $\kappa=10^{-5}$, $\sigma_j=1/5+j/10$, $B_j=1/2+\sigma_j$, and $C_j=1+\sigma_j$. All asserted gains of the source's cycle must exceed $1/10$ after derivative losses. For example,

$$\frac{18}{100}-2\kappa>\frac{17}{100}>\frac{1}{10},\qquad \sigma_j-3\kappa\ge\frac{1}{5}-3\kappa>\frac{17}{100},$$
$$\frac{1}{2}-2\kappa>\frac{1}{10},\qquad \frac{9}{10}-4\kappa>\frac{1}{10}.$$

These rational inequalities certify arithmetic slack once the functional estimates are supplied. They do not establish those functional estimates. This is a useful separation for learned proof search: a

model may propose an ordering or an exponent assignment, while exact arithmetic and analytic lemmas discharge different obligations.

2.8 Localization and smooth force extension

Proposition 2.11 (Localization without a divergence defect)**.** *Suppose A, Be_θ and p are smooth in Cartesian coordinates, put $u = \nabla \times A + Be_\theta$, and assume B is independent of θ. Let c be a smooth axisymmetric scalar. Set*

$$\widetilde{u} = \nabla \times (cA) + cBe_\theta, \qquad \widetilde{p} = cp, \qquad a = \nabla c \times A.$$

Then $\widetilde{u} = cu + a$ and $\nabla \cdot \widetilde{u} = 0$. Its residual is exactly

$$\begin{aligned}\mathcal{R}_\nu(\widetilde{u}, cp) = c\mathcal{R}_\nu(u,p) + (\partial_t c - \nu\Delta c)u - 2\nu\sum_j (\partial_j c)\partial_j u + p\nabla c \\ + (c^2 - c)(u\cdot\nabla)u + c(u\cdot\nabla c)u \\ + \partial_t a - \nu\Delta a + c(u\cdot\nabla)a + (a\cdot\nabla)(cu) + (a\cdot\nabla)a.\end{aligned}$$

Proof. The curl product rule gives $\widetilde{u} = cu + a$. Divergence of a curl is zero; the other term has divergence $r^{-1}\partial_\theta(cB) = 0$ away from the axis and hence everywhere by Cartesian smoothness. Expand $\partial_t(cu)$, $\Delta(cu)$, and $\nabla(cp)$ by the product rule. Finally, $(cu\cdot\nabla)(cu) = c^2(u\cdot\nabla)u + c(u\cdot\nabla c)u$. Adding the three advection terms containing a yields the formula. □

The displayed expression explains why a spatial cutoff requires endpoint regularity of the actual potentials as well as the velocity. On a region where $c = 1$ the extra terms vanish. On its transition region every term involving the uncut fields must remain smooth through the terminal time. A proof that estimates only $c\mathcal{R}_\nu(u,p)$ has omitted real terms.

Lemma 2.12 (Realization of terminal force jets)**.** *Let $f(x,t)$ be smooth for $t < T$, supported in a fixed compact set $K \subset \mathbb{R}^d$. Suppose every $\partial_x^\alpha \partial_t^j f$ has a uniform limit $\partial_x^\alpha F_j(x)$ as $t \uparrow T$, where $F_j \in C_c^\infty(\mathbb{R}^d)$ are supported in K. Then f has a smooth extension to $t \geq T$ supported in $K \times [T, T+1]$ on that side. If it already vanishes near an initial time, the combined extension retains that property.*

Proof. Choose a smooth cutoff χ, one near zero and zero for arguments at least one, and for $s = t - T \geq 0$ define

$$f_+(x, T+s) = \sum_{j=0}^{\infty} \chi(b_j s)\frac{s^j}{j!}F_j(x).$$

For fixed α, m, the product rule and $s \leq b_j^{-1}$ on the summand's support give a bound

$$C_{j,m} b_j^{m-j} \|F_j\|_{C^{|\alpha|}}.$$

Fix $b_0 = 1$. For $j \geq 1$, choose increasing $b_j \geq 1$ so that this is at most 2^{-j} whenever $|\alpha| + m \leq \lfloor j/2 \rfloor$. Each step imposes finitely many constraints with $m - j < 0$, so the choice exists. Every differentiated tail converges uniformly. At $s = 0$, the mth time derivative of a fixed summand is F_m when $j = m$ and zero otherwise. Uniform differentiated convergence gives the desired jets.

Uniform convergence of the original mixed derivatives makes their limits compatible: pass to the limit in the fundamental theorem of calculus along a coordinate segment, and then along a time segment. The two pieces therefore glue smoothly at T. All summands retain support in K, and vanish for $s \geq 1$; uniform convergence of every derivative also proves smoothness at that support boundary. □

Flatness at a single singular point is only part of this lemma's hypotheses. One must additionally control all terminal derivatives away from that point, including the cutoff transition region. Conversely, vanishing of every jet does not force a smooth function to vanish near the point: the function e^{-1/s^2} for $s > 0$, extended by zero for $s \le 0$, is the standard counterexample. Substituting "analytic" for "smooth" would change the problem and invalidate this extension method.

2.9 Comparison, quantitative residual checks, and proof strength

An explicit singular solution rules out a global competitor only after uniqueness has been established in the competitor's class. The whole-space class in the Clay problem permits no arbitrary decay assumption on the pressure. That is why a periodic energy calculation is not by itself a complete proof of alternative (C). The source's Lemma 10.5 handles this point using a pressure-gradient argument and expanding spatial cutoffs; its hypotheses must be preserved in any proof adapter [26].

The following separate estimate is useful for the project's computable fluid examples and for any learned approximate field.

Proposition 2.13 (Residual-to-solution estimate on the flat torus)**.** *Let u, v be smooth divergence-free periodic vector fields on $[0,T]$, and let periodic p, q satisfy*

$$\mathcal{R}_\nu(u,p) = f, \qquad \mathcal{R}_\nu(v,q) = f + r.$$

Write $w = v - u$ and $K(t) = \|\nabla u(t)\|_{L^\infty,\mathrm{op}}$. If K is integrable, then

$$\|w(t)\|_2 \le e^{\int_0^t K(a)\,da}\|w(0)\|_2 + \int_0^t e^{\int_s^t K(a)\,da}\|r(s)\|_2\,ds.$$

In particular, equal data and $r = 0$ imply $v = u$.

Proof. Subtract the equations in the form

$$\partial_t w + (v\cdot\nabla)w + (w\cdot\nabla)u - \nu\Delta w + \nabla(q-p) = r.$$

Periodic integration and zero divergence remove the transport and pressure pairings. Thus

$$\tfrac{1}{2}(\|w\|_2^2)' + \nu\|\nabla w\|_2^2 \le K\|w\|_2^2 + \|r\|_2\|w\|_2.$$

With $y_\delta = (\|w\|_2^2 + \delta^2)^{1/2}$ one has $y_\delta' \le K y_\delta + \|r\|_2$. Multiplication by $e^{-\int_0^t K}$ and integration prove the bound after $\delta \downarrow 0$. □

This inequality gives a precise role for a quantum or classical learner: it may produce v, but a certificate needs its divergence defect, momentum residual, initial error, and an appropriate stability coefficient. The estimate does not remain effective near blowup merely because the sampled residual is small. A pointwise grid evaluation does not bound the continuum L^2 residual without an interpolation or enclosure argument, and $\int K$ can become large. Numerical fitting and a posteriori proof are different stages of the same workflow.

The same issue explains why nearby classical results are not interchangeable. Nonuniqueness in a weak class does not give blowup of a smooth solution; a modified averaged nonlinearity is not the original Navier–Stokes nonlinearity; and rough forcing is not smooth forcing. These distinctions are visible respectively in the settings of Buckmaster–Vicol, Tao, and Albritton–Brué–Colombo [1, 6, 30].

2.10 Viscosity, periodization, and preservation of quantifiers

Proposition 2.14 (Viscosity conversion)**.** *If $\mathcal{R}_1(u,p) = f$ and $\nabla \cdot u = 0$, then for $\nu > 0$*

$$u_\nu(x,t) = \sqrt{\nu}\, u(x/\sqrt{\nu}, t), \quad p_\nu(x,t) = \nu p(x/\sqrt{\nu}, t), \quad f_\nu(x,t) = \sqrt{\nu}\, f(x/\sqrt{\nu}, t)$$

satisfy $\mathcal{R}_\nu(u_\nu, p_\nu) = f_\nu$ and zero divergence. The singular time is unchanged. In three dimensions,

$$\|u_\nu(t)\|_2^2 = \nu^{5/2}\|u(t)\|_2^2, \qquad \nu\|\nabla u_\nu(t)\|_2^2 = \nu^{5/2}\|\nabla u(t)\|_2^2.$$

Proof. Put $y = x/\sqrt{\nu}$. Each term of the momentum equation acquires a factor $\sqrt{\nu}$: two spatial derivatives contribute ν^{-1}, one contributes $\nu^{-1/2}$, and the amplitudes contribute the stated powers. The divergence equals $(\nabla_y \cdot u)(y,t)$. The Jacobian of $x = \sqrt{\nu}y$ is $\nu^{3/2}$, while the velocity square contributes ν and the gradient square contributes one. This proves both norm identities. The time argument was not changed. □

Proposition 2.15 (Disjoint-support periodization)**.** *Suppose smooth whole-space fields u, p, f solve Navier–Stokes with viscosity ν and have common spatial support in a compact set $K \Subset (-1/2, 1/2)^3$. Then their sums over integer translates define smooth periodic fields solving the same equation, with the same viscosity.*

Proof. The sums are locally finite. The distinct translated compact supports have positive separation, so at each point a neighborhood intersects at most one support, unless the fields vanish there. Differential operators commute with the sums. In the advection term every mixed product of different translates vanishes, leaving the sum of the individual advection terms. The same argument gives zero divergence and retains periodicity of both velocity and pressure. □

For a larger compact support one first uses the parabolic scaling

$$\widetilde{u}(x,t) = \lambda u(\lambda x, \lambda^2(t - t_0)), \quad \widetilde{p}(x,t) = \lambda^2 p(\lambda x, \lambda^2(t - t_0)), \quad \widetilde{f}(x,t) = \lambda^3 f(\lambda x, \lambda^2(t - t_0)).$$

Choosing λ large puts the support inside the cube. If the original fields vanish on an initial interval, they extend smoothly by zero before t_0. For original singular time one, $t_0 = 1 - \lambda^{-2}$ keeps the new singular time equal to one. These conditions explain why periodizing an arbitrary spatially decaying field would not suffice: overlapping copies would create additional nonlinear interactions.

2.11 Formal proof provenance and what this extension adds

The pinned upstream revision is

`f9e8bc5b38b6e212696e8a30e3e91517af887bbd`.

Its toolchain is Lean 4.34.0-rc2, with mathlib revision

`85e3a25e006c35636f0e53b0e9296caca2685bc0`.

The project's earlier selected Lean declarations use a different toolchain. They cannot be merged by silently substituting one compiler for the other. An isolated verification script preserves the upstream pins.

The final upstream declarations are

```
NavierStokes.Comparator.navier_stokes_breakdown_R3
NavierStokes.Comparator.navier_stokes_breakdown_periodic.
```

Their types quantify over every positive viscosity and retain the exact initial-data, forcing, regularity, energy, and periodicity predicates. The reference challenge modules intentionally contain unproved challenge statements. They are kept separate from the solution imports. A source scan that merely counts every occurrence of `sorry` in the whole repository would therefore misdescribe the solution's dependency boundary. Conversely, a source scan finding no such token cannot replace kernel checking or comparison of the definitions with the intended mathematics.

The inspection found 816 files under the Navier–Stokes source directory and 609 local modules in the final solution's import closure. Selected modules were read in detail; all 609 were not manually reviewed. The upstream metadata reports only the standard dependencies `propext`, `Classical.choice`, and `Quot.sound` for the final results. The accompanying verification ledger distinguishes that upstream report from any fresh build and independent Comparator execution performed in this extension [25].

In this review the exact compiler distribution was downloaded and its published SHA-256 checksum matched. Dependency cache retrieval then failed with HTTP 429 responses after partial decompression. A direct attempt to elaborate the final module stopped at missing local import artifacts. Consequently this pass did not complete a fresh build of the final upstream theorems, and did not run the independent Comparator certificate check. The source audit found no challenge import and no unproved-placeholder token in the 609-module local solution closure, after removing comments and string literals. That lexical result has the narrower meaning stated here; it is not a substitute for the uncompleted kernel verification.

The extension's new contribution is an explicit mathematical organization: quadratic residual calculus; exact information-loss identities; norm scaling that separates energy from regularity; a forbidden-word estimate that explains a common analytic domain; all-order realization with correct quantifiers; potential-based localization; and a quantitative residual comparison for learned fields. Chapter 3 then develops the arithmetic, exterior-power, and filtered-algebra connections, while Chapter 4 treats quantum representations and learning.

These are rigorous bridges because each has a specified map or estimate. They do not identify a finite surface-code complex with the continuum Navier–Stokes equation, or turn a learned function into an existence theorem without a stability certificate. They also preserve the released theorem's force. Removing that force, imposing a different physical constitutive law, or deriving the same singular flow from a microscopic quantum state would be a further theorem with additional analytic content.

Chapter 3

Orders, complexes and the algebra of a fluid singularity

Evidence status. This chapter gives original, explicit proofs of the finite algebraic and differential identities used to interpret the released construction. The construction of the singular Navier–Stokes solution remains the external theorem of OpenAI [26], reviewed in Chapter 2. The real auxiliary torus, the physical three-dimensional fluid, a Riemann surface and a quantum register are different objects. Every identification below specifies which structure it preserves.

3.1 What a useful reformulation must preserve

The central opportunity is to reorganize the proof into mathematical interfaces that the inherited project already studies: integral endomorphisms, exterior powers, complexes, quadratic tensor maps, moment determinants and filtered operator algebras. This is more informative than attaching algebraic terminology to a picture of a vortex. It also creates obligations. A change of notation must preserve the domain, positive viscosity, zero initial data, compact smooth forcing, bounded kinetic energy and the claimed divergence of velocity. An algebraic identity must not silently replace an analytic estimate that is uniform in the concentrating scale.

In the released proof the physical equation is on $\mathbb{R}^3$, with a subsequent periodic construction on $\mathbb{T}^3$. The auxiliary $\mathbb{T}^2$ carries oscillatory labels and is evaluated along a prescribed phase map. It is neither the physical domain nor a quantum phase space. A surface-algebra model can encode incidence, permitted paths and local tensor types. To obtain an actual fluid discretization one must additionally choose a metric, a volume form, boundary conditions, constitutive parameters and approximation operators. Those data will remain visible throughout this chapter.

3.2 The de Rham complex behind incompressible motion

Let M be an oriented flat three-dimensional manifold without boundary, or use fields on $\mathbb{R}^3$ for which the integrations below have no boundary terms. Write $\alpha = u^\flat$ for the velocity one-form, δ for the metric codifferential and $\Delta_H = d\delta + \delta d$ for the nonnegative Hodge Laplacian. With the usual Euclidean sign convention, Δ_H acts componentwise as minus the coordinate Laplacian. Thus incompressibility is $\delta\alpha = 0$.

Proposition 3.1 (The differential-form momentum equation)**.** *For a smooth divergence-free velocity, the usual forced Navier–Stokes equation is equivalent to*

$$\partial_t \alpha + \iota_u d\alpha + \nu \Delta_H \alpha + d\pi = f^\flat, \qquad \delta\alpha = 0, \qquad \pi = p + \tfrac{1}{2}|u|^2.$$

Its vorticity two-form $\omega = d\alpha$ satisfies

$$\partial_t \omega + \mathcal{L}_u \omega + \nu \Delta_H \omega = df^\flat.$$

Proof. The ith component of $\iota_u d\alpha$ is $\sum_j u_j(\partial_j u_i - \partial_i u_j) = (u \cdot \nabla)u_i - \partial_i(|u|^2/2)$. Adding $d\pi$ restores the advective and pressure terms. The sign of Δ_H gives $-\nu\Delta u$ on the left. Apply d to this equation. It commutes with ∂_t and with $d\delta + \delta d$ because $d^2 = 0$. Cartan's identity gives $d\iota_u\omega = \mathcal{L}_u\omega - \iota_u d\omega = \mathcal{L}_u\omega$. The pressure disappears by $d^2\pi = 0$. □

This places exterior powers in their proper role. The momentum flux $u \otimes u$ is symmetric; vorticity is alternating. The two occupy Sym^2 and $\bigwedge^2$ tensor types, respectively. They interact through differential operators and metric contractions. Exchanging their symmetry types would change the equation.

Proposition 3.2 (Energy and the work of a differential)**.** *Under the hypotheses just stated,*

$$\frac{d}{dt}\frac{1}{2}\|u\|_2^2 + \nu\|\nabla u\|_2^2 = \langle f, u\rangle_{L^2}.$$

In particular neither pressure nor transport contributes to the total kinetic-energy derivative.

Proof. The pressure contribution is $\langle dp, \alpha\rangle = \langle p, \delta\alpha\rangle = 0$. Pointwise, $\langle \iota_u d\alpha, \alpha\rangle = d\alpha(u, u) = 0$, and $\langle d(|u|^2/2), \alpha\rangle$ integrates to zero. Integration by parts gives $\langle -\Delta u, u\rangle = \|\nabla u\|_2^2$. Finally $\langle \partial_t u, u\rangle = \frac{1}{2}\partial_t\|u\|_2^2$. □

The proposition controls an integral. It does not bound $\|u\|_\infty$. This matters especially in the newly released construction: concentration makes the energetic cost of a large speed small. The scale computation in Chapter 2 is therefore compatible with this identity, rather than a violation of it. Nor does dissipation imply that every individual mode decreases: transport can redistribute energy between scales while its total work vanishes.

On a curved manifold one must specify which viscous operator is intended. The Hodge Laplacian, the rough Laplacian and the divergence of a symmetric strain tensor differ by curvature contributions under standard conventions. A topological surface model cannot choose between them. The flat torus examples below fix the Hodge convention explicitly. Their familiar three-dimensional torus drawing is a visualization of periodic coordinates, not an assertion that the induced metric of the embedded doughnut is flat.

3.3 A quadratic order occurs in the actual construction

Section 6 of OpenAI [26] chooses the integer matrix

$$J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \qquad A = J - 4I = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its arithmetic controls two different derivative scales on an auxiliary torus. This is a direct meeting point with the orders developed earlier in the monograph.

Proposition 3.3 (The integral quadratic-order representation)**.** *One has* $A^2 = 2I$, $J^2 - 8J + 14I = 0$ *and* $\det J = 14$. *The map*

$$\rho : \mathbb{Z}[\sqrt{2}] \longrightarrow M_2(\mathbb{Z}), \qquad a + b\sqrt{2} \longmapsto aI + bA$$

is an injective unital ring homomorphism, and

$$\det \rho(a + b\sqrt{2}) = a^2 - 2b^2.$$

Consequently $J = \rho(4 + \sqrt{2})$ *and* $\det(J^k) = 14^k$ *for every integer* $k \geq 0$.

Proof. Matrix multiplication gives $A^2 = 2I$. Hence $(aI + bA)(cI + dA) = (ac + 2bd)I + (ad + bc)A$, the multiplication law of the quadratic order. Its matrix is $\left(\begin{smallmatrix} a-b & b \\ b & a+b \end{smallmatrix}\right)$, whose determinant is $a^2 - 2b^2$. If it is the zero matrix its off-diagonal entry gives $b = 0$, then $a = 0$, proving injectivity. Expanding $(J - 4I)^2 = 2I$ gives the polynomial identity. Determinants are multiplicative, including the case $k = 0$. □

The determinant is also the scalar by which the top exterior power acts: $\bigwedge^2 J = 14 \operatorname{id}_{\bigwedge^2 \mathbb{Z}^2}$. Thus the same alternating-tensor construction that gave the determinant and Cauchy–Binet earlier now measures a covering multiplicity. Since 14 is not a unit of $\mathbb{Z}$, this is a covering endomorphism, not an invertible integral change of basis.

Proposition 3.4 (Covering multiplicity and exact averaging)**.** *The map* $\pi_k : \mathbb{R}^2/\mathbb{Z}^2 \to \mathbb{R}^2/\mathbb{Z}^2$ *given by* $[y] \mapsto [J^k y]$ *is a surjective covering with* 14^k *points in each fiber. For normalized Haar measure,*

$$\int_{\mathbb{T}^2} F(J^k y)\, dy = \int_{\mathbb{T}^2} F(y)\, dy$$

for every continuous F.

Proof. Invertibility over $\mathbb{R}$ gives surjectivity. A fiber is a translate of $(J^{-k}\mathbb{Z}^2)/\mathbb{Z}^2$, isomorphic to $\mathbb{Z}^2/J^k\mathbb{Z}^2$. Integer row and column operations reduce the latter quotient to Smith normal form, whose order is $|\det J^k| = 14^k$. Locally the real linear map is a diffeomorphism, proving the covering assertion. For a character $e_n(y) = e^{2\pi i n \cdot y}$, pullback gives $e_{(J^k)^T n}$. Because J^k is invertible, this character has nonzero integral exactly when $n = 0$. Finite linear combinations of characters satisfy the integral identity. Uniform approximation by trigonometric polynomials extends it to continuous functions. □

This argument does not invoke a long-time ergodic limit. It is an exact identity for a finite covering. In the released proof one uses a common auxiliary torus for overlapping dyadic bands, keeping the difference of covering levels bounded. Its derivative factors are then uniformly bounded even though the absolute covering level grows. Omitting this distinction would lose the uniformity needed later in the estimates.

Proposition 3.5 (Why this is not a complex elliptic-curve endomorphism)**.** *There is no real linear complex structure* C *on* $\mathbb{R}^2$, *meaning* $C^2 = -I$, *that commutes with* J.

Proof. The real symmetric matrix J has distinct real eigenvalues $4 - \sqrt{2}$ and $4 + \sqrt{2}$. Every commuting map preserves each of its one-dimensional real eigenspaces. On each, C is multiplication by a real scalar c, so C^2 is multiplication by $c^2 \geq 0$ and cannot be -1. □

Therefore the auxiliary covering has an arithmetic description by a quadratic order, but cannot be advertised as a holomorphic endomorphism of one fixed complex elliptic curve. The fact that its underlying real manifold is a torus supplies too little structure for that conclusion. This distinction also prevents an unsupported identification with the elliptic-curve groups used by cryptography.

3.4 The number-field norm supplies an analytic estimate

Set $b = \sqrt{2} - 1$, $v_r = (1, -b)$ and $v_t = (b, 1)$. Direct calculation gives $Jv_r = (4 - \sqrt{2})v_r$ and $Jv_t = (4 + \sqrt{2})v_t$. Their irrational directions are useful because no nonzero integer frequency is perpendicular to either one, and the failure of perpendicularity has a quantitative lower bound.

Lemma 3.6 (A quadratic small-divisor bound)**.** *There is an explicit constant* $c = (1 + (1 + \sqrt{2})^2)^{-1/2} > 0$ *such that*

$$|v_r \cdot n| \geq c/|n|_2, \qquad |v_t \cdot n| \geq c/|n|_2 \quad (n \in \mathbb{Z}^2 \setminus \{0\}).$$

Proof. For the first expression put $a = n_1 + n_2$ and $d = -n_2$. Then $v_r \cdot n = a + d\sqrt{2}$. Its algebraic norm $a^2 - 2d^2$ is a nonzero integer: otherwise irrationality of $\sqrt{2}$ would force $a = d = 0$, and hence $n = 0$. Its absolute value is at least one. The conjugate is $n_1 + (1 + \sqrt{2})n_2$, of absolute value at most $c^{-1}|n|_2$ by Cauchy–Schwarz. Dividing the norm by this bound proves the first assertion. The other direction is $(n_2 - n_1) + n_1\sqrt{2}$, whose conjugate has the same coefficient norm. □

Proposition 3.7 (A tame inverse of a torus directional derivative)**.** *Let* v *be either direction above, and let* F *be a smooth zero-mean function on* $\mathbb{T}^2$*. There is a unique smooth zero-mean* ϕ *satisfying* $v \cdot \nabla\phi = F$*, namely*

$$\phi(y) = \sum_{n \neq 0} \frac{\widehat{F}(n)}{2\pi i\, v \cdot n} e^{2\pi i n \cdot y}.$$

For every real s *there is* C_s *with* $\|\phi\|_{H^s} \leq C_s \|F\|_{H^{s+1}}$*. For every nonnegative integer* m*, one also has* $\|\phi\|_{C^m} \leq C_m \|F\|_{C^{m+4}}$*.*

Proof. The lemma bounds the inverse multiplier by $C|n|$. Squaring and summing the weighted Fourier coefficients proves the Sobolev estimate. For the uniform derivative estimate, integration by parts bounds a Fourier coefficient of a C^{m+4} function by $C(1 + |n|)^{-m-4}\|F\|_{C^{m+4}}$. Multiplying by the m derivative factors and the inverse divisor leaves $C(1 + |n|)^{-3}$, summable in two dimensions. This proves uniform convergence of all required derivative series. Applying the same argument for arbitrarily large m proves smoothness. Uniqueness follows because every nonzero Fourier coefficient of a solution of the homogeneous equation vanishes, and the zero coefficient was prescribed. □

Here an integral norm identity yields a finite derivative loss, which is then paid for using smoothness at every fixed order. That is a genuine analytic contribution of the arithmetic structure. It is not an estimate uniform in derivative order, and it would not justify infinite-order flatness from a fixed finite regularity assumption. In Section 8 of the released proof, repeated integration by parts along such directions makes compact-support cutoff errors smaller than any specified power. The number of input derivatives increases with the requested power.

3.5 Symmetric tensors, covariance and the stress correction

For a real velocity perturbation w averaged by a positive probability measure, $\langle w \otimes w \rangle$ is positive semidefinite: for every a its quadratic form is $\langle (a \cdot w)^2 \rangle \geq 0$. It belongs to the symmetric tensor representation. A single $w \otimes w$ has rank at most one, so all its two-by-two minors vanish. The determinantal variety describes this algebraic rank constraint, but averaging rank-one matrices can produce a full-rank matrix. Rank should not be transported through an average without proof.

The released construction needs two specified radial cross-flux components rather than an arbitrary covariance matrix. If b_+ and b_- have disjoint auxiliary supports, their mixed products vanish pointwise. Writing

$$\mathcal{C}(w) = \langle w_r(w_\theta, w_z) \rangle , \qquad H = [\mathcal{C}(b_+) \mid \mathcal{C}(b_-)],$$

gives $\mathcal{C}(a_+b_+ + a_-b_-) = H(a_+^2, a_-^2)^T$. A cross-flux vector itself is not a positive semidefinite matrix; positivity concerns the coefficients of this two-ray representation.

Proposition 3.8 (Positive realization and a signed tangent inverse)**.** *Let $H \in \mathrm{GL}_2(\mathbb{R})$, let $s \in \mathbb{R}^2$ satisfy $y = H^{-1}s \in (0,\infty)^2$, and put $a_i = \sqrt{y_i}$. For $C(a) = H(a_1^2, a_2^2)^T$, define*

$$L_a\sigma = \mathrm{diag}(2a_1, 2a_2)^{-1} H^{-1}\sigma.$$

Then $C(a) = s$, $DC(a)L_a\sigma = \sigma$, and

$$C(a + L_a\sigma) = s + \sigma + H\big((L_a\sigma)_1^2, (L_a\sigma)_2^2\big)^T.$$

The last term is generally nonzero.

Proof. The first identity substitutes $a_i^2 = y_i$. Differentiating the coordinate squares gives $DC(a) = H\,\mathrm{diag}(2a_1, 2a_2)$, whose displayed right inverse exists since both amplitudes are positive. Expanding $(a_i + h_i)^2 = a_i^2 + 2a_ih_i + h_i^2$ proves the final identity. □

This is the same distinction between tangent information and an exact finite deformation that appeared in the varieties-of-complexes chapter. The determinant of H proves a local algebraic inverse exists. The PDE proof additionally needs a lower bound for that determinant, control of derivatives of H^{-1}, and flat edge factors that permit division by the leading amplitudes. It keeps those amplitudes fixed while applying signed increments and accounts for the quadratic remainder at each subsequent step. Replacing this procedure by the claim that every target stress is a positive covariance would be false.

The Schur-functor viewpoint is useful here without overreach. In characteristic zero, the flip on $V \otimes V$ yields the complementary projectors $(I+\mathrm{flip})/2$ and $(I-\mathrm{flip})/2$ onto symmetric and alternating tensors. The velocity covariance uses the first; the differential two-form uses the second. Higher tensor lifts use Young symmetries to remove redundant coordinates. Integral straightening theory controls the corresponding bases before denominators are inverted. None of these representation decompositions by itself supplies the scale estimates of the analytic construction.

3.6 Moment determinants and exact compatibility

Compactly supported primitives have obstructions. If $(\partial_r + e/r)\phi = g$ with ϕ compactly supported away from zero, multiplication by r^e and integration gives $\int r^e g\, dr = 0$. Conversely, for real e and $g \in C_c^\infty((0,\infty))$ satisfying that moment condition, $\phi(r) = r^{-e}\int_0^r s^e g(s)\, ds$ is compactly supported and solves the equation. This elementary exactness criterion is the prototype for the weighted radial inverses in the proof. Torus-dependent coefficients require the additional phase shifts and flat remainders described there.

Proposition 3.9 (Independence of ordered moment functionals)**.** *Let $\alpha_1, \dots, \alpha_m$ be distinct real numbers, and let β_j be nonzero, nonnegative smooth bumps supported inside disjoint ordered intervals $0 < I_1 < \cdots < I_m$. Then*

$$B_{ij} = \int_{I_j} x^{\alpha_i} \beta_j(x)\, dx$$

defines an invertible matrix. Smooth variation over a compact parameter set gives bounded inverse derivatives provided all the hypotheses, including distinctness of the exponents, persist there.

Proof. A nonzero linear combination of m distinct powers has at most $m-1$ distinct positive zeros. For induction divide by the smallest power. If the resulting expression is a nonzero constant, it has no zeros. Otherwise its nonzero derivative is a combination of at most $m-1$ distinct powers and has at most $m-2$ zeros by the induction hypothesis; Rolle's theorem then proves the assertion. Hence $\det[x_j^{\alpha_i}]$ never vanishes for $0 < x_1 < \cdots < x_m$. It has constant sign on that connected chamber. Multilinearity and Fubini give

$$\det B = \int_{I_1 \times \cdots \times I_m} \det[x_j^{\alpha_i}] \prod_j \beta_j(x_j)\, dx_1 \cdots dx_m.$$

The integrand has a fixed sign and is nonzero on a set of positive measure. Thus the determinant cannot vanish. Compactness bounds it away from zero for a compact family satisfying the hypotheses. The adjugate formula and repeated differentiation give the inverse-derivative bounds. □

For geometrically rescaled copies of one bump, the matrix becomes a Vandermonde matrix after row normalization. This connects the proof to the same alternating determinants and interpolation combinatorics used earlier. The quantitative limitation is crucial: if two exponents coalesce, the inverse can diverge. The parameter called λ in the source is fixed before the high radial frequency is chosen. Sending it to zero inside an argument that needs a bounded inverse would alter the proof.

Lemma 3.10 (A controlled quadratic moment correction)**.** *Suppose B is invertible, Q is bilinear, $\|B^{-1}\| \le \beta$, $\|Q\| \le \kappa$, and $\|d\| \le d_0$. If $8\beta^2\kappa d_0 \le 1$, the equation $Bc + Q(c,c) = d$ has a unique solution in the closed ball $\|c\| \le 2\beta d_0$.*

Proof. Set $r = 2\beta d_0$ and $T(c) = B^{-1}(d - Q(c,c))$. For $\|c\| \le r$, $\|T(c)\| \le \beta d_0 + \beta\kappa r^2 \le 3r/4 \le r$. For two points in the ball, bilinearity gives $\|T(c) - T(c')\| \le 2\beta\kappa r\|c - c'\| \le \frac{1}{2}\|c - c'\|$. The contraction theorem gives the unique fixed point in that ball. The fixed-point equation is exactly the required moment equation. □

The estimate is a local solvability statement. It does not establish that a whole family of discrepancies is small; that must be proved from the profile construction and its order of parameter choices. This is the appropriate division of labor between algebraic elimination and analysis.

3.7 From varieties of complexes to compatible fluid discretization

Let a finite oriented cell decomposition of a closed surface have real cochains C^0, C^1, C^2 and coboundary matrices $d_0 : C^0 \to C^1$, $d_1 : C^1 \to C^2$ with $d_1 d_0 = 0$. The matrix entries are signed incidence numbers. These matrices give a point of the affine variety of complexes defined by the entries of $d_1 d_0$. Additional rank conditions select the strata studied earlier. The underlying incidence relation is exact over the integers; a floating-point residual is only a computational check of its implementation.

Choose symmetric positive definite mass matrices M_0, M_1, M_2. Define

$$\delta_1 = M_0^{-1} d_0^T M_1, \qquad \delta_2 = M_1^{-1} d_1^T M_2, \qquad L_1 = d_0\delta_1 + \delta_2 d_1.$$

These are the chosen discrete adjoints and Hodge operator. Geometry enters through the M_j, not through the relation $d_1 d_0 = 0$ alone. Stable approximation requires additional commuting projection and norm estimates; finite element exterior calculus makes these requirements precise [3, 4].

Proposition 3.11 (Discrete Hodge energy and incompressible subspace)**.** *For every $x \in C^1$,*

$$x^T M_1 L_1 x = \|\delta_1 x\|^2_{M_0} + \|d_1 x\|^2_{M_2} \geq 0.$$

Moreover L_1 preserves $\ker \delta_1$*. A velocity of the form $\delta_2 \psi$ belongs to this kernel.*

Proof. The definitions identify d_0 and δ_1, and d_1 and δ_2, as adjoints in the indicated inner products. Move each adjoint across the pairing to obtain the sum of squares. Taking adjoints of $d_1 d_0 = 0$ gives $\delta_1 \delta_2 = 0$. Therefore $\delta_1 L_1 = (\delta_1 d_0)\delta_1$, which is zero on $\ker \delta_1$. The last assertion is the same zero-composition identity. □

An arbitrary algebraic change of basis preserves the complex equation. It preserves the numerical energy only if the metric is transported as well. If $x' = Tx$, then the corresponding matrix is $M_1' = T^{-T} M_1 T^{-1}$. Retaining the identity metric after a badly conditioned change of basis can change coercivity constants arbitrarily. This repeats, in the fluid setting, the distinction between homological equivalence and quantitative spectral control.

Theorem 3.12 (An energy-certified finite learned evolution)**.** *Work in orthonormal coordinates on a finite-dimensional divergence-free subspace. Let $K(a)$ be a locally Lipschitz skew-symmetric matrix, L a symmetric positive semidefinite matrix, and $B_\theta(a)$ a locally Lipschitz matrix. For continuous forcing $g(t)$, consider*

$$\dot{a} = -K(a)a - \nu La - B_\theta(a)^T B_\theta(a)a + g(t), \qquad \nu > 0.$$

It satisfies

$$\tfrac{1}{2}\frac{d}{dt}|a|^2 = -\nu a^T La - |B_\theta(a)a|^2 + a \cdot g(t).$$

Every solution exists for all finite positive times, provided the displayed coefficients are defined and locally Lipschitz on the whole finite-dimensional space.

Proof. For a skew matrix $a^T K(a)a$ equals its own negative, hence is zero. The factorized term has work $|B_\theta(a)a|^2$. This proves the identity. Young's inequality yields $\frac{d}{dt}|a|^2 \leq |a|^2 + |g(t)|^2$. Gronwall gives a finite bound on every finite interval because g is continuous. A locally Lipschitz finite-dimensional ODE can be continued while its solution stays in a compact subset of its domain. The bound therefore rules out finite-time escape and proves global continuation. □

The theorem is a useful design rule for a learned closure, including one whose matrix entries are produced by quantum expectation values. It is also a useful warning: every fixed finite model can be globally well behaved while the continuum family develops unbounded pointwise velocity. The equivalence constant between a grid energy norm and the maximum grid value typically diverges under refinement. Global continuation of each finite ODE cannot be passed to the continuum without uniform regularity and consistency estimates.

The dissipative ansatz is deliberately restrictive. True interscale transfer can include backscatter, while $-B_\theta^T B_\theta$ never injects resolved energy. A trained model in this class must be assessed for that modeling bias. The algebraic certificate proves stability of its own evolution, not fidelity to the exact coarse dynamics.

3.8 Gentle algebras, path relations and quantitative interfaces

The project uses gentle surface algebras to organize permitted local paths and relations. A compatible use here is to let vertices label local field types or mesh patches, arrows label permitted transfers, and quadratic relations encode compositions known to vanish. The equation $d^2 = 0$ is one such zero composition. The derivative-block relation in the Volterra construction, discussed in Chapter 2, is another actual example. Counting admissible words then bounds how often a derivative-losing operator can appear.

One must verify the gentle conditions for the chosen quiver and relations. A free algebra with a single relation $b^2 = 0$ is not automatically gentle: allowed continuations at its other generator can violate the required local uniqueness. Nor does an arbitrary triangulated surface incidence quiver become gentle merely because it is drawn on a surface. The valid general statement is that a representation of a specified path algebra with relations encodes the corresponding exact operator identities. Calling it gentle requires the extra combinatorial checks already described in the earlier chapters.

An associated graded construction is likewise conditional on a specified filtration. If classes $\mathcal{M}_\alpha$ obey $\mathcal{M}_\alpha\mathcal{M}_\beta \subseteq \mathcal{M}_{\alpha+\beta}$, they track orders under multiplication. If a derivative maps $\mathcal{M}_\alpha$ into $\mathcal{M}_{\alpha-\kappa}$, its loss must be included in a correction cycle. Logarithmic factors and flat shell weights are part of the analytic definition. A finitely generated semigroup algebra, a flat toric degeneration, normality or a Schur-module decomposition does not follow solely from this notation. Such claims require the corresponding algebraic hypotheses and proofs.

Standard monomial theory is valuable when one introduces an actual finite coordinate ring of rank constraints, tensor symmetries or moment minors. It provides bases, relations and algebraic elimination. The Navier–Stokes construction further needs positivity margins, compact support and all-order differentiability. These conditions are not encoded by an affine scheme over $\mathbb{C}$. In particular positivity is real semialgebraic or analytic data, and flat nonzero functions cannot be recovered from a convergent analytic Taylor series at their flat endpoint.

3.9 Determinants, spectra and what the reformulation adds

For a finite positive operator L with eigenvalues λ_j, the identities

$$\det(sI + L) = \prod_j (s + \lambda_j), \qquad \frac{d}{ds} \log \det(sI + L) = \operatorname{tr}(sI + L)^{-1}$$

hold for $s > 0$ by diagonalization and ordinary differentiation. Exterior powers and Schur constructions organize coefficients of the determinant. A continuum determinant requires a specified regularization and domain; a spectral zeta function requires convergence or analytic continuation. No identification with an arithmetic L-function, its zeros, or the inherited GRH program follows from these finite identities.

The improvement supplied by this extension is therefore explicit: the integral torus covering is expressed as an order action; its norm becomes a small-divisor estimate; covariance realization becomes a quadratic map with a retained remainder; moment matching becomes a determinant and contraction problem; incompressibility becomes a complex relation; and learned finite models receive a verifiable energy interface. The released proof's analytic construction is preserved rather than replaced. A stronger theorem, such as a result for unforced Navier–Stokes or a quantitative algorithm producing its full singular field, would require additional work beyond these reformulations.

Chapter 4

Quantum information, exact reduction and learned fluid models

Evidence status. The reduction, channel and scalar Carleman statements below are proved explicitly. The computational companion evaluates finite classical simulations of the stated models. Quantum algorithms cited here have their own input, error and readout assumptions. No quantum speedup, trained fluid ToricGT model, or numerical reproduction of the released singular solution is claimed.

4.1 Information reduction is a map, not an autonomous dynamics

The microscopic-to-macroscopic hierarchy in Chapter 1 is naturally expressed by maps that retain observables. Examples include taking a partial trace, a one-particle marginal, a velocity moment, an angular average, or a projection onto a finite set of modes. These maps can be exact even when the subsequent closed evolution is approximate. The distinction is elementary: knowing a projected state need not determine the derivative of that state.

Proposition 4.1 (The obstruction to an exact instantaneous closure)**.** *Consider $\dot{x} = Ax + Bz$ on finite-dimensional vector spaces. If $By \neq 0$ for some hidden vector y, no function $F(x)$ can satisfy $F(x) = Ax + Bz$ for every pair (x, z).*

Proof. For fixed x, compare the hidden states $z = 0$ and $z = y$. The proposed equality gives both $F(x) = Ax$ and $F(x) = Ax + By$, contradicting $By \neq 0$. □

The statement does not rule out a closure on a specified invariant manifold, a conditional expectation under a prescribed ensemble, or an approximation with controlled error. It rules out omitting those hypotheses. It applies equally to a classical neural network and a quantum learned function: an expressive predictor cannot make an insufficient input state mathematically sufficient.

Theorem 4.2 (Exact memory from a linear hidden subsystem)**.** *Let A, B, C, D be constant compatible finite matrices and consider*

$$\dot{x} = Ax + Bz, \qquad \dot{z} = Cx + Dz.$$

Then

$$\dot{x}(t) = Ax(t) + Be^{tD}z(0) + \int_0^t Be^{(t-s)D}Cx(s)\,ds.$$

Conversely, any continuously differentiable x satisfying this equation, together with $z(t) = e^{tD}z(0) + \int_0^t e^{(t-s)D}Cx(s)\,ds$, satisfies the original system.

Proof. Multiply the hidden equation by e^{-tD} and differentiate the product. Integration gives the displayed expression for z. Substitution gives the equation for x. In the converse direction, differentiate the finite matrix-exponential integral using the fundamental theorem of calculus. This recovers $\dot{z} = Cx + Dz$, and substitution gives the other equation. □

This is a finite, fully explicit version of the memory mechanism associated with projection methods [24]. The memory kernel is $K(t) = Be^{tD}C$, and the hidden initial condition supplies a separate term. Dropping that term is not justified simply by calling it noise. Its distribution and correlations must be supplied if a stochastic description is intended.

For example, with scalar equations $\dot{x} = -2x + z$, $\dot{z} = x - 3z$, the exact kernel is e^{-3t}. Even when $z(0) = 0$, replacing the reduced equation by $\dot{x} = -2x$ discards a nonzero convolution. A fast-relaxation approximation can instead use $z \approx x/3$, but its error depends on the relative time scales and initial layer. Exact elimination and time-scale approximation are different steps.

Proposition 4.3 (Discrete memory and the resolvent determinant)**.** *For $x_{n+1} = Ax_n + Bz_n$, $z_{n+1} = Cx_n + Dz_n$,*

$$z_n = D^n z_0 + \sum_{j=0}^{n-1} D^{n-1-j} C x_j,$$

and substitution gives the exact resolved recurrence. If $sI - D$ is invertible, the determinant of the full block resolvent satisfies

$$\det \begin{pmatrix} sI - A & -B \\ -C & sI - D \end{pmatrix} = \det(sI - D)\det\big(sI - A - B(sI - D)^{-1}C\big).$$

Proof. The hidden-state formula follows by induction, with the sum empty at $n = 0$. Multiplying by D and adding Cx_n advances the sum to $n + 1$. For the determinant, multiply the block matrix on the left by $\left(\begin{smallmatrix} I & B(sI-D)^{-1} \\ 0 & I \end{smallmatrix}\right)$, whose determinant is one. The upper-right block becomes zero and the upper-left block becomes the displayed Schur complement. The determinant of the resulting block triangular matrix is the product of its diagonal-block determinants. □

This links the project's determinant calculus to an actual information-reduction operation. The frequency dependence of the Schur complement is the transform of a memory kernel. Replacing it by a constant is a modeling approximation. The existence of its inverse does not provide a uniform condition-number bound, especially near a spectral pole.

4.2 Coarse-graining and the unresolved stress

For a normalized nonnegative averaging kernel, write $\bar{u} = \langle u \rangle$ and $R = \langle u \otimes u \rangle - \bar{u} \otimes \bar{u}$. The quadratic nonlinearity ensures that the equation for $\bar{u}$ contains a contribution involving R. A projection that retains only $\bar{u}$ has discarded information relevant to the momentum equation.

Proposition 4.4 (Covariance as the exact quadratic defect)**.** *For a probability average with finite second moments (at a fixed observation point, when the average is spatially local),*

$$R = \langle (u - \bar{u}) \otimes (u - \bar{u}) \rangle \succeq 0, \quad \operatorname{tr} R = \langle |u|^2 \rangle - |\bar{u}|^2.$$

If a linear filter commutes with spatial derivatives, filtering $\nabla \cdot (u \otimes u)$ gives $\nabla \cdot (\bar{u} \otimes \bar{u}) + \nabla \cdot R$.

Proof. Expand the centered product, use $\langle u \rangle = \bar{u}$ and $\langle 1 \rangle = 1$, and take traces. Testing the matrix against any vector gives an average of a square. The final identity follows by linearity, the definition of R, and the assumed derivative commutation. □

For a convolution evaluated at x, the centered integrand in this proof is $u(y) - \bar{u}(x)$, averaged over y. It is not $u(y) - \bar{u}(y)$ for a general non-idempotent filter.

A sharp Fourier truncation need not be a positive averaging kernel. Its quadratic defect is still defined, but the positive-semidefinite conclusion requires the probability-average hypothesis. Similarly, a filter near a wall need not commute with differentiation. Those are concrete places where otherwise plausible information-theoretic arguments fail if their maps are left unspecified.

In the released blowup proof, oscillatory perturbations generate selected averaged quadratic fluxes. The authors construct those fluxes and bound their errors; they do not infer them from an assumed generic turbulence closure. The covariance inverse in Proposition 3.8 captures one finite algebraic component of that construction. Its analytic realization retains the carrier phase, exact curl correction, support separation and derivative estimates.

4.3 A quantum observable can have the same closure obstruction

For a finite quantum system with density matrix ρ and Hermitian observable O, the retained datum $\mathrm{tr}(\rho O)$ is only one linear functional of the state. Even noiseless knowledge of it need not determine its future derivative.

Example 4.5 (Identical observed value, opposite future derivative)**.** Take $\hbar = 1$, Hamiltonian $H = X$ and observable $O = Z$, with Pauli matrices in the conventions of the earlier quantum chapters. The states $\rho_\pm = (I \pm Y)/2$ both have $\mathrm{tr}(\rho_\pm Z) = 0$. Unitary evolution gives

$$\frac{d}{dt}\langle Z \rangle = i\langle [X, Z] \rangle = 2\langle Y \rangle,$$

so their derivatives are $+2$ and -2.

The commutator is obtained by direct multiplication, $[X, Z] = -2iY$. Thus the example requires no interpretive claim about measurement disturbance. The retained statistic simply omits a degree of freedom that matters dynamically. A reduced density matrix can likewise acquire memory when a subsystem is coupled to an environment. A time-homogeneous Markov semigroup requires additional assumptions; it does not follow from partial trace alone.

The same point controls proposed quantum information explanations of viscosity. Closed unitary evolution preserves von Neumann entropy. Effective irreversible evolution arises after a specified reduction and limiting or modeling assumptions. A completely positive trace-preserving map provides a consistent finite open-system operation, but an arbitrary such map is not a derivation of the Navier–Stokes constitutive law.

4.4 An explicit heat channel on a periodic surface grid

Let the position register have orthonormal basis $\{|j, k\rangle : j, k \in \mathbb{Z}/L\mathbb{Z}\}$. Let T_x, T_y be the unitary permutations that translate by one grid point. For $0 \le r \le 1/4$ define

$$\Phi_r(\rho) = (1 - 4r)\rho + r(T_x \rho T_x^\dagger + T_x^\dagger \rho T_x + T_y \rho T_y^\dagger + T_y^\dagger \rho T_y).$$

Proposition 4.6 (A trace-preserving quantum diffusion step)**.** *The map Φ_r is completely positive, trace preserving and unital. Its diagonal probabilities obey*

$$p'_{jk} = (1-4r)p_{jk} + r(p_{j-1,k} + p_{j+1,k} + p_{j,k-1} + p_{j,k+1}).$$

They remain nonnegative, sum to one, and their Shannon entropy cannot decrease.

Proof. Use Kraus operators $\sqrt{1-4r}I$, $\sqrt{r}T_x$, $\sqrt{r}T_x^\dagger$, $\sqrt{r}T_y$, $\sqrt{r}T_y^\dagger$. Their adjoint products sum to I, as do the reversed products. This proves complete positivity, trace preservation and unitality. Taking diagonal entries gives the stencil. It is a convex combination of translations of the probability vector. The function $-s\log s$ is concave on $[0,1]$, with its continuous value zero at zero. Jensen's inequality at each site, followed by summation and translation invariance, proves the entropy assertion. □

Proposition 4.7 (Discrete heat energy and eigenvalues)**.** *Let L be a positive integer, $h>0$ and $\nu \geq 0$. On the periodic $L \times L$ grid, define $\Delta_h p = h^{-2}(T_x + T_x^{-1} + T_y + T_y^{-1} - 4I)p$. Then*

$$-\langle p, \Delta_h p\rangle = h^{-2}\sum_{j,k}((p_{j+1,k}-p_{jk})^2 + (p_{j,k+1}-p_{jk})^2).$$

The Fourier mode $e^{2\pi i(aj+bk)/L}$ has eigenvalue $-4h^{-2}[\sin^2(\pi a/L) + \sin^2(\pi b/L)]$. The heat equation $\dot{p} = \nu\Delta_h p$ therefore contracts squared Euclidean norm and preserves the constant mode.

Proof. Expand each squared difference and reindex periodic sums to obtain the quadratic identity. Translation acts on the Fourier mode by $e^{\pm 2\pi i a/L}$ or $e^{\pm 2\pi i b/L}$. The formula follows from $2\cos\theta - 2 = -4\sin^2(\theta/2)$. Differentiating $\frac{1}{2}\|p\|^2$ along the heat equation and using the first identity proves contraction. All shifts fix the constant vector. □

With $r = \nu\Delta t/h^2$, the channel gives an explicit heat step. Its positivity requires the displayed time-step bound. The continuous semigroup is the limit of these steps, equivalently the exponential of the random-translation generator. This is an exact finite quantum-channel interpretation of scalar diffusion. The browser computes it classically on a 16×16 grid. The position space has dimension 256, representable by eight qubits, but the computation is not an eight-qubit hardware experiment or a solution of nonlinear Navier–Stokes. A signed velocity field is also not a probability distribution; any encoding of it must preserve signs and normalization by an additional construction.

4.5 What the Madelung transformation does and does not derive

On a simply connected open region where a smooth scalar wave function does not vanish, write $\psi = \sqrt{\rho}\, e^{iS/\hbar}$. For a particle of mass $m>0$ in a real potential V, substitute this into $i\hbar\partial_t\psi = -(\hbar^2/2m)\Delta\psi + V\psi$.

Proposition 4.8 (The local quantum hydrodynamic equations)**.** *With $v = \nabla S/m$ and $Q = -(\hbar^2/2m)\Delta\sqrt{\rho}/\sqrt{\rho}$, one obtains*

$$\partial_t\rho + \nabla\cdot(\rho v) = 0, \qquad \partial_t S + \frac{|\nabla S|^2}{2m} + V + Q = 0.$$

Consequently $m(\partial_t v + (v\cdot\nabla)v) = -\nabla(V+Q)$ and $\nabla\times v = 0$ in that region.

Proof. Differentiate the amplitude-phase product twice. Its Laplacian is

$$e^{iS/\hbar}\left[\Delta\sqrt{\rho}+\frac{2i}{\hbar}\nabla\sqrt{\rho}\cdot\nabla S+\frac{i}{\hbar}\sqrt{\rho}\,\Delta S-\frac{1}{\hbar^2}\sqrt{\rho}\,|\nabla S|^2\right].$$

Equate imaginary parts and multiply by $2\sqrt{\rho}$ to obtain continuity. Divide the real part by $\sqrt{\rho}$ to obtain the second equation. Its gradient gives the velocity equation because the Hessian of S is symmetric, so $\nabla(|v|^2/2)=(v\cdot\nabla)v$. The curl of a gradient is zero. □

There is no viscous term in this calculation. At zeros of ψ the amplitude-phase chart fails; around a closed loop with a single-valued nonzero wave function the phase winding can give circulation $2\pi\hbar n/m$. One scalar phase on a nonvanishing simply connected patch therefore cannot represent an arbitrary smooth vortical incompressible velocity. More general spinor, mixed-state or multi-component encodings may represent richer currents, but their dynamical correspondence must be separately derived. The Born–Oppenheimer and kinetic limits reviewed earlier are distinct routes, not consequences of the scalar Madelung identity alone.

4.6 Carleman lifting and the symmetric algebra

Consider a finite quadratic system $\dot{x}=Ax+B(x\otimes x)+c$, with $x\in\mathbb{R}^n$. Differentiating a tensor power gives

$$\frac{d}{dt}x^{\otimes m}=\sum_{j=1}^{m}x^{\otimes(j-1)}\otimes\dot{x}\otimes x^{\otimes(m-j)}.$$

Substitution couples tensor degrees $m-1,m,m+1$. Thus the infinite collection of powers evolves linearly in a larger space. This is an identity along an existing solution, not a conclusion that the infinite system has a bounded solution for all time.

All tensors $x^{\otimes m}$ are symmetric. In commuting coordinates the degree-m monomials form a basis of dimension $\binom{n+m-1}{m}$ rather than n^m. The symmetric algebra and its standard monomials therefore provide an exact reduction of redundant tensor coordinates. Differentiation acts as a derivation on this algebra. Exterior powers have a different role: they encode determinants and rank constraints, not powers of one commuting vector, whose repeated wedge is zero.

Proposition 4.9 (An exact scalar truncation calculation)**.** *Let $N\geq 1$ be an integer, $\alpha>0$, $\beta\geq 0$, $x_0>0$, and consider $\dot{x}=-\alpha x+\beta x^2$. Put $r=\beta x_0/\alpha$ and $z(t)=r(1-e^{-\alpha t})$. Before the denominator vanishes,*

$$x(t)=\frac{x_0e^{-\alpha t}}{1-z(t)}.$$

Truncate the lifted equations $\dot{y}_m=-m\alpha y_m+m\beta y_{m+1}$ at $m=N$ by setting $y_{N+1}=0$ and $y_m(0)=x_0^m$. Their first component is

$$y_1^{(N)}(t)=x_0e^{-\alpha t}\sum_{j=0}^{N-1}z(t)^j,\qquad x(t)-y_1^{(N)}(t)=x_0e^{-\alpha t}\frac{z(t)^N}{1-z(t)}.$$

If $r<1$, the error is bounded by $x_0r^N/(1-r)$ for all $t\geq 0$. If $r>1$, the nonlinear solution blows up at $t_*=-\alpha^{-1}\log(1-1/r)$; the finite lifted system does not remove that singularity.

Proof. For the exact equation, $w = 1/x$ satisfies $\dot{w} = \alpha w - \beta$. Solving this linear equation yields the displayed formula. The truncated triangular system is solved by downward variation of constants. More explicitly its mth component is

$$y_m^{(N)}(t) = x_0^m e^{-m\alpha t} \sum_{j=0}^{N-m} \binom{m+j-1}{j} z(t)^j.$$

Differentiate this expression, use $z' = \beta x_0 e^{-\alpha t}$ and $j\binom{m+j-1}{j} = m\binom{m+j-1}{j-1}$, and check the initial values. Uniqueness for the finite linear system proves the formula. At $m = 1$ it is the geometric partial sum. Subtracting from the exact rational expression gives the remainder. The bound follows from $0 \le z(t) \le r < 1$. When $r > 1$, the denominator first vanishes at the stated positive time, while the numerator remains positive. □

This calculation is a small, rigorous model of the convergence issue faced by quantum linearization. The finite lift is globally solvable even when it no longer approximates the nonlinear solution. It follows that solving a truncated lifted system, classically or quantum mechanically, cannot establish continuum regularity without a truncation estimate valid in the relevant regime.

4.7 What the quantum-algorithm literature actually supplies

Liu et al. [19] analyze a dissipative quadratic ODE algorithm with a parameter measuring nonlinear and forcing strength relative to linear dissipation. Their stated favorable regime requires $R < 1$, together with sparse access, state preparation, error and solution-norm assumptions. The final output is a normalized quantum state encoding a vector; obtaining a full classical velocity field has an additional readout cost. Their numerical Burgers examples outside the sufficient regime are evidence about those examples, not a theorem covering every Reynolds number. The retrieved arXiv source is version 4, dated 7 July 2026. That correction adds $\|F_0\| \le \|F_2\|$ to the relevant convergence statements. It does not repair a separate eigenvalue-to-norm inference in the printed Lemma 1, which must not be imported into this project.

Proposition 4.10 (A counterexample to eigenvalue-only norm decay)**.** *For the real quadratic system $\dot{u} = F_1 u + F_2(u \otimes u) + F_0$, choose*

$$F_1 = \begin{pmatrix} -1 & 4 \\ 0 & -2 \end{pmatrix}, \quad F_2 = \begin{pmatrix} 1/100 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad F_0 = \begin{pmatrix} 1/1000 \\ 0 \end{pmatrix}, \quad u(0) = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

The matrix F_1 is diagonalizable with strictly negative eigenvalues. In the Euclidean and induced spectral norms, the source's parameter is $R = 3\sqrt{2}/500 < 1$, and $\|F_0\| \le \|F_2\|$. Nevertheless

$$\frac{d}{dt}\|u(t)\|^2\Big|_{t=0} = \frac{1007}{2000} > 0.$$

In particular these hypotheses do not imply $\|u(t)\| < \|u(0)\|$ for every positive sufficiently small t.

Proof. The distinct eigenvalues are $-1, -2$, the three relevant norms are $\|u(0)\| = 1/\sqrt{2}$, $\|F_2\| = 1/100$, $\|F_0\| = 1/1000$, and substitution gives the stated R. Direct evaluation gives $\dot{u}(0) = (3007/2000, -1)^T$. Therefore $2u(0)^T\dot{u}(0) = 1007/2000$. The bound asserted by the source's equation (4.2), with spectral real part -1, would instead be $-1 + 3\sqrt{2}/500 < 0$. The exact positive derivative contradicts that bound and yields an interval of initial norm increase by differentiability. □

The problem is that $u^*(F_1 + F_1^*)u$ is controlled by the largest eigenvalue of the Hermitian part, not by twice the largest real part of the eigenvalues of an arbitrary nonnormal matrix. The source specifies Euclidean and spectral norms and does not first transform the system into an eigenbasis. Its scalar rescaling leaves F_1 unchanged. The counterexample concerns that particular printed inference; it is not a claim that every application or a suitably repaired quantum algorithm fails.

Proposition 4.11 (A sufficient dissipative norm hypothesis)**.** *Suppose $(F_1 + F_1^*)/2 \preceq -\alpha I$ with $\alpha > 0$, $\|F_2\| \le \beta$, and $\|F_0(t)\| \le \gamma$, with continuous forcing. If a radius $a > 0$ satisfies $\beta a^2 - \alpha a + \gamma < 0$, every solution starting in $\|u(0)\| \le a$ remains in that ball and continues for all positive times.*

Proof. At a nonzero state of radius r, differentiation and Cauchy–Schwarz give $\dot r \le -\alpha r + \beta r^2 + \gamma$. At the boundary $r = a$ the right side is strictly negative. Continuity makes it negative on a sufficiently small annulus about that radius, so a trajectory cannot first cross that annulus outwards. A boundary initial state immediately moves inward by the same inequality. Thus the ball is invariant. The polynomial vector field is locally Lipschitz in the state, and continuity of the forcing on each finite time interval and boundedness of the trajectory give global finite-dimensional continuation. □

This repaired condition is stronger than spectral stability and is exactly the kind of additional metric information demanded throughout the monograph. A weighted inner product can sometimes establish such a condition, but its conditioning and the transformed nonlinear norm must then enter the quantitative and algorithmic bounds. We use only the independently proved estimates here, not the unrepaired general norm-decay lemma.

Non-dissipative extensions also need hypotheses. Theorem 1.1 of the reviewed version of Wu et al. [40] assumes a diagonalizable F_1, eigenvalues with nonpositive real parts, a uniform solution bound $\|x(t)\|_1 \le \mu$, and the strictly positive resonance gap

$$\Delta = \inf_{k,\, m \in \mathbb{N}^n,\, |m| \ge 2} \left| \lambda_k - \sum_j m_j \lambda_j \right| > 0.$$

Here $\mathbb{N}$ includes zero. Its stated estimate uses $R_r = 4e\mu\kappa_1(W)\|F_2\|_1/\Delta$; geometric convergence requires $R_r < 1$. Merely asserting that each individual resonance expression is nonzero does not establish a positive infimum over infinitely many expressions. The condition also has a particularly restrictive consequence for real systems.

Proposition 4.12 (Real nonresonance excludes neutral oscillatory modes)**.** *Let F_1 be a real matrix, with all eigenvalues in the closed left half-plane. If no eigenvalue equals a sum of at least two eigenvalues, allowing repetition, then every eigenvalue has strictly negative real part.*

Proof. A zero eigenvalue violates the hypothesis because $0 = 0 + 0$. A nonzero purely imaginary eigenvalue $i\omega$ of a real matrix has the conjugate eigenvalue $-i\omega$. Then $i\omega = 2i\omega + (-i\omega)$ is a sum with three terms, again violating the hypothesis. These are all possibilities on the imaginary axis, so the nonpositive real parts are strictly negative. □

Consequently, the title of that paper cannot be used to assert a theorem for arbitrary real conservative or neutral fluid systems. Its numerical examples and its precise theorem assumptions must be assessed separately. This observation does not refute its nonresonant estimate; it identifies the real-system domain in which that estimate could apply. We do not use the source's stronger estimate as a dependency of the released singularity proof. A direct estimate adequate for the present finite comparison is as follows.

Proposition 4.13 (A norm-controlled short-time Carleman estimate)**.** *Let $N \geq 1$ be an integer. Suppose $\dot{x} = F_1 x + F_2 x^{\otimes 2}$ on $[0, T]$, with $\|x(t)\| \leq M$. Use compatible Euclidean tensor norms or compatible ℓ^1 tensor norms, and assume $\|e^{tF_1}\| \leq 1$ for $t \geq 0$. For the degree-N Carleman truncation initialized by $y_j(0) = x(0)^{\otimes j}$,*

$$\|x(t) - y_1(t)\| \leq M\big(M\|F_2\|t\big)^N, \qquad 0 \leq t \leq T.$$

In particular the right side converges geometrically in N when $M\|F_2\|T < 1$.

Proof. The diagonal degree-j generator is $A_{j,j} = \sum_{i=0}^{j-1} I^{\otimes i} \otimes F_1 \otimes I^{\otimes(j-1-i)}$. Its semigroup is $(e^{tF_1})^{\otimes j}$ and is contractive under the stated norms. The adjacent block is the analogous sum with F_2 replacing one input slot; its norm is at most $j\|F_2\|$. Write $\eta_j = x^{\otimes j} - y_j$. All initial errors vanish, and the sole terminal forcing is $A_{N,N+1}x^{\otimes(N+1)}$. Repeated variation of constants expresses η_1 as an N-fold ordered integral. The integrand norm is at most $N!\|F_2\|^N M^{N+1}$. The ordered simplex has volume $t^N/N!$, giving the bound. □

Diagonalizability with nonpositive spectral real parts alone does not imply contractivity in the original norm. If $F_1 = W\Lambda W^{-1}$, one can apply the proposition to $z = W^{-1}x$, with the transformed nonlinearity $\widetilde{F}_2 = W^{-1}F_2(W \otimes W)$ and a bound on z. Returning to x costs $\|W\|$. This makes the conditioning visible instead of silently discarding it. Semisimplicity of neutral eigenspaces also differs from claiming that every eigenvalue has geometric multiplicity one. These distinctions correct misleading shortcuts in the preliminary estimates of the reviewed preprint without importing them into our argument.

For our project the required interface is a bound for the particular lifted operator and initial state, on the requested time interval, with its dependence on discretization made explicit.

The September 2026 end-to-end study by Jennings et al. [16] is especially relevant. It accounts for Carleman truncation, time stepping, conditioning, unitary block construction and observable extraction in a lattice-Boltzmann approach. Its shifted formulation and discrete lifting address specific earlier bottlenecks, but its conclusions concern selected observables and parameter regimes. The paper reports numerical convergence and nonconvergence behavior under changing truncation order, and distinguishes proven bounds from fitted scaling. We retain that distinction. Neither its quantum encoding nor its finite Reynolds-number experiments are a proof of the released three-dimensional singularity theorem.

Two further approaches deserve separate names. Variational quantum methods can represent nonlinear functionals using several copies of a state and a chosen ansatz [22]. Their practical result depends on optimization, ansatz approximation, measurement variance and implementation costs. Tensor-network methods such as Gourianov et al. [14] can exploit interscale correlations on classical computers; their compression depends on the ranks needed by the particular flow. A tensor-network simulation is not a quantum-device execution. Both can be useful research tools without claiming a universal exponential speedup.

For every proposed fluid algorithm, the complete cost should specify at least the spatial approximation, time discretization, nonlinear lift or ansatz error, state preparation, success probability, conditioning and requested observable. Normalizing a large vector does not make its small physically relevant component free to read out. A statement polynomial in the number of qubits can still have a large or exponential dependence on physical accuracy, evolution time or a hidden condition number.

4.8 Exact finite repairs for collision block encodings

The detailed algorithm of Jennings et al. [16] makes it possible to inspect the finite maps underlying its collision encoding. In the reviewed arXiv version 2, Appendix D, equation (D56), gives a purported unitary for an equality map, while Appendix I, equation (I65), gives a matrix dilation without its necessary contraction normalization. We supply explicit finite replacements. These proofs concern the two interfaces; they do not establish Carleman convergence, a total resource bound, or quantum advantage for the complete fluid algorithm.

Let $\mathcal{H}$ have orthonormal basis $\{|a\rangle : a \in \{0,1\}^n\}$, with $n \geq 1$. Padding a position register to a power of two permits this notation. Define

$$S = \sum_a |a,0\rangle\langle a,a|, \qquad \Pi_{\mathrm{eq}} = \sum_a |a,a\rangle\langle a,a|, \qquad \Pi_0 = \sum_a |a,0\rangle\langle a,0|.$$

The register 0 is the all-zero bit string. The map S retains equal input positions and resets the second register on that subspace.

Proposition 4.14 (Equality map as a partial isometry)**.** *One has $S^\dagger S = \Pi_{\mathrm{eq}}$, $SS^\dagger = \Pi_0$, and $\|S\| = 1$. The expression*

$$B = S \otimes I + (S - I) \otimes X$$

on two position registers and a flag qubit is not unitary.

Proof. Multiply the two finite sums for S and $S^\dagger$. Orthonormality eliminates every cross term, giving the two projectors. Hence S is isometric on the equality subspace and zero on its orthogonal complement, which proves the norm statement. For any nonzero a,

$$B|a,a,0\rangle = |a,0,0\rangle + |a,0,1\rangle - |a,a,1\rangle.$$

The three output vectors are pairwise orthogonal. Thus a unit input has output squared norm three, contradicting unitarity. □

This counterexample applies to the expression printed in (D56), with its stated square embedding $S = \sum_a |a,0\rangle\langle a,a|$. It is not an obstruction to constructing an equality block encoding.

Proposition 4.15 (Reversible equality encoding and its success probability)**.** *Define $U|a,b\rangle = |a, b \operatorname{xor} a\rangle$. Then U is unitary and $\Pi_0 U = S$. A flag measurement can realize this block with success probability $\|Sx\|^2$ on any normalized input $x \in \mathcal{H} \otimes \mathcal{H}$.*

Proof. The basis permutation U is its own inverse. Applying Π_0 to its output retains exactly the inputs with $b = a$, proving $\Pi_0 U = S$. For an explicit flag, define the reversible basis permutation

$$V|a,c,z\rangle = |a,c,z \operatorname{xor} \mathbf{1}_{c\neq 0}\rangle, \qquad W = V(U \otimes I).$$

Both factors are unitary. Starting the flag in $|0\rangle$ and retaining outcome zero gives

$$(I \otimes \langle 0|)W(x \otimes |0\rangle) = Sx.$$

The Born rule yields success probability $\|Sx\|^2$. Writing $x = \sum_{a,b} x_{ab}|a,b\rangle$ makes this probability $\sum_a |x_{aa}|^2$. When it is positive, the conditional output is $Sx/\|Sx\|$; when it is zero, that outcome never occurs. □

No copying of an arbitrary unknown quantum state occurs here. The XOR operation copies a basis label into a zero register, and on superpositions acts coherently, generally producing correlations. Applied to $x \otimes x$, the equality branch contains squared amplitudes, but preparing two copies already presupposes repeated access to the state preparation. Its success probability remains part of the operation.

Proposition 4.16 (Contraction-normalized rectangular dilation)**.** *Let $A : \mathcal{H}_{\mathrm{in}} \to \mathcal{H}_{\mathrm{out}}$ be a finite complex matrix and let $\alpha > 0$ satisfy $\alpha \geq \|A\|$. Put $C = A/\alpha$ and*

$$D_{\mathrm{out}} = (I - CC^{\dagger})^{1/2}, \qquad D_{\mathrm{in}} = (I - C^{\dagger}C)^{1/2}.$$

Then

$$\mathcal{U}_C = \begin{pmatrix} C & D_{\mathrm{out}} \\ D_{\mathrm{in}} & -C^{\dagger} \end{pmatrix} : \mathcal{H}_{\mathrm{in}} \oplus \mathcal{H}_{\mathrm{out}} \longrightarrow \mathcal{H}_{\mathrm{out}} \oplus \mathcal{H}_{\mathrm{in}}$$

is unitary. On normalized input $(x, 0)$, projection onto the first output summand succeeds with probability $\|Ax\|^2/\alpha^2$ and, when successful, produces $Ax/\|Ax\|$. Conversely, a block of a unitary representing A/α requires $\alpha \geq \|A\|$.

Proof. Take a singular-value decomposition $C = L\Sigma R^{\dagger}$. Every singular value s_j lies in $[0, 1]$. The positive square roots therefore exist, and the SVD gives $CD_{\mathrm{in}} = D_{\mathrm{out}}C$: on each paired singular direction both sides multiply by $s_j\sqrt{1 - s_j^2}$, and both vanish on the unpaired directions. Consequently

$$\mathcal{U}_C\mathcal{U}_C^{\dagger} = \begin{pmatrix} CC^{\dagger} + D_{\mathrm{out}}^2 & CD_{\mathrm{in}} - D_{\mathrm{out}}C \\ D_{\mathrm{in}}C^{\dagger} - C^{\dagger}D_{\mathrm{out}} & D_{\mathrm{in}}^2 + C^{\dagger}C \end{pmatrix} = I.$$

The same calculation gives $\mathcal{U}_C^{\dagger}\mathcal{U}_C = I$. Its action on $(x, 0)$ is $(Cx, D_{\mathrm{in}}x)$, proving the measurement formulas. Finally, the norm of a compression of a unitary is at most one, so any such block must satisfy $\|A/\alpha\| \leq 1$. □

Thus (I65) must use the contraction Σ_2/α, including the same scaling inside both square roots. The reviewed paper's equation (D58a) gives $\alpha = 6/\bar{\tau}^{\star}$ for its $D = 2$ collision matrix. A normalization cannot be removed merely because the requested output is eventually renormalized: it determines the success amplitude and enters the resources of the larger block-encoding algorithm.

Proposition 4.17 (Linear instrument, conditional normalization)**.** *The successful and unsuccessful branches of the preceding dilation define linear completely positive maps*

$$\rho \longmapsto C\rho C^{\dagger}, \qquad \rho \longmapsto D_{\mathrm{in}}\rho D_{\mathrm{in}}.$$

Their total trace equals $\operatorname{tr}\rho$*. Normalizing the successful branch does not turn it into deterministic nonlinear quantum evolution.*

Proof. Each branch has a single Kraus operator and is completely positive: after any ancillary extension it sends a positive operator to a conjugate of that operator. Since $C^{\dagger}C + D_{\mathrm{in}}^2 = I$, cyclicity of trace proves the trace identity. The conditional state is divided by $\operatorname{tr}(C\rho C^{\dagger})$ only after the corresponding random outcome occurs. Discarding that probability would change the physical operation. □

4.9 A mathematically constrained quantum ToricGT extension

The previously implemented quantum ToricGT was trained and checked on tokenization-path data. Its weights cannot be presented as a validated fluid model. The reusable component is its architecture: incidence-aware vertex and edge tokens, local quantum features, classical aggregation and an explicit certificate after prediction. This section specifies a fluid adaptation and proves the structural certificate independently of training.

For a surface mesh, tokens should include oriented incidence, the chosen mass matrices, local coordinates or invariant metric features, boundary tags, resolved velocity and any retained memory state. A surface's embedding alone is insufficient. Gauge-like changes of cochain basis should transform the fields and metric together. Under mesh relabeling, the local circuit input and classical aggregation must obey a stated equivariance rule; otherwise predictions can depend on arbitrary vertex numbering.

Let $|\psi_\theta(x)\rangle = U_\theta(x)|0\rangle$ be a normalized finite circuit and let $q_j(x) = \langle\psi_\theta(x)|O_j|\psi_\theta(x)\rangle$ for Hermitian O_j with $\|O_j\| \le 1$. Then $q_j \in [-1, 1]$. A classical readout can construct a matrix $B_\theta(x)$ and a skew matrix $K_\theta(x) = \frac{1}{2}(A_\theta(x) - A_\theta(x)^T)$. In divergence-free coordinates, use the vector field of Theorem 3.12. The resulting dissipation identity is true for every choice of circuit parameters; it does not rely on the optimizer finding a good model.

Proposition 4.18 (A parameterized positive constitutive coefficient)**.** *Let $0 < \nu_{\min} < \nu_{\max}$ and let $\ell_\theta(x)$ be any finite real readout of circuit features. Then*

$$\nu_\theta(x) = \nu_{\min} + (\nu_{\max} - \nu_{\min})\frac{1}{1 + e^{-\ell_\theta(x)}}$$

lies strictly between the two bounds. If the readout is locally Lipschitz, using this coefficient in a factorized finite viscosity preserves the energy sign in Theorem 3.12.

Proof. The exponential is positive, so the logistic factor is strictly between zero and one. Its derivative is bounded. Composition with a locally Lipschitz readout is locally Lipschitz. Multiplication of a squared norm by a positive scalar preserves nonnegativity, giving the asserted sign. □

Proposition 4.19 (The circuit derivative and a nonlinear loss)**.** *Suppose one parameter occurs in one gate $e^{-i\theta P/2}$ with $P = P^*$ and $P^2 = I$. For any expectation $q(\theta)$ measured after the full circuit,*

$$q'(\theta) = \tfrac{1}{2}[q(\theta + \pi/2) - q(\theta - \pi/2)].$$

For a differentiable loss $L(q_1, \dots, q_m)$, its derivative is the chain-rule sum $\sum_j (\partial L/\partial q_j) q_j'$. Shifting the entire nonlinear loss is not generally the same operation.

Proof. Expand the gate as $\cos(\theta/2)I - i\sin(\theta/2)P$. The expectation is of the form $a + b\cos\theta + c\sin\theta$ after the fixed preceding and following operations are absorbed into the state and observable. Differentiation and the two shifted trigonometric identities prove the first assertion. The second is ordinary multivariable differentiation. For example, if $q = \cos\theta$ and $L = q^2$, shifting L by $\pm\pi/2$ gives equal values and an erroneous zero, whereas $L' = -\sin 2\theta$ in general. □

Shared parameters in multiple gates require a sum of gate-occurrence derivatives. The previous ToricGT experiments already distinguish this chain rule from an incorrect whole-loss shift. For a fluid extension it is equally essential: a conservation or energy penalty is a nonlinear loss and must be differentiated correctly.

4.9.1 A concrete training and validation specification

A future trained fluid ToricGT should be assessed against the same dynamical problem solved by its classical baselines. Data must identify viscosity, forcing, domain, mesh, time step, reference solver and truncation error. Split by trajectories and physical parameter families, not by randomly mixing nearby snapshots of the same trajectory. Include independently held-out meshes and resolutions. Compare with a classical model of similar trainable size, a frozen-circuit model, the physical baseline without learned closure and an exact manufactured solution.

Report resolved trajectory error, derivative or stress error where available, divergence, energy balance, pressure residual and total wall time including data transfer and readout. A projection or repair can make a constraint pass even when a predictor is inaccurate, so pre-repair and post-repair results must be distinguished. A small training residual is not a certified PDE residual unless its sampling, quadrature and interpolation errors have also been bounded.

Memory is a plausible place to use learning: approximate a stable auxiliary realization of a kernel, with enough state to distinguish histories that share the same instantaneous resolved velocity. This is more defensible than asking a memoryless network to invert an information-losing projection. For a linear stable hidden subsystem, Theorem 4.2 supplies a reference target and a complete exact comparison. For a nonlinear flow, the corresponding error and stability hypotheses require new analysis.

4.10 The executable examples and their proof boundaries

The new laboratory separates four tasks. The Taylor–Green field is an exact two-dimensional periodic Navier–Stokes solution with known pressure and decay, used to test derivatives and normalization. The integer covering panel evaluates the actual order matrix from the released proof. The circuit panel computes a two-qubit state and uses its bounded features to select a positive heat-channel parameter. The concentrating-core panel displays the released scaling exponents, without claiming to compute the source's fully corrected singular field.

The heat circuit is an untrained parameterized example. It is not a fluid-trained ToricGT, and the finite heat channel does not include the nonlinear momentum term. The code records actual finite invariants and numerical discrepancies. The companion's selected Lean declarations formalize energy, projection, finite memory and order identities, with precise component mappings; they do not certify every browser operation or every theorem in these chapters.

Taken together, these examples make information reduction operational. One can see which quantities are preserved by a complex, which are lost under projection, which reappear as a memory kernel or a quadratic stress, and which remain controlled by a positive quantum-channel construction. The released Navier–Stokes proof then occupies its proper place: a continuum construction whose all-order analytic estimates establish a forced singularity, and whose algebraic components can be understood and checked within this wider framework.

Appendix A

Fluid verification, reproducibility and proof dependencies

A.1 What was checked and what each check means

The fluid extension has three distinct verification layers. The manuscript gives explicit proofs of the displayed finite statements and analytical interfaces. The local Lean package checks selected statements with precisely typed hypotheses. The executable examples compare independently implemented finite calculations. None of these layers is interchangeable with the complete upstream continuum development.

The additional file `extension/lean/FluidInterfaces.lean` contains 34 theorem declarations. It was elaborated successfully by Lean 4.24.0 against the same pinned mathlib revision as the earlier companion,

`f897ebcf72cd16f89ab4577d0c826cd14afaafc7`.

The check returned exit code zero and no warnings. Every declaration has a printed type and an axiom report. The only reported foundations are `propext`, `Classical.choice` and `Quot.sound`; no admitted theorem or project-specific axiom is introduced. The verified source SHA-256 is

`9ac752e3450fd07f9fb23cb4d0283ecee1b70d7a3188ece1cdcff5d8b66934b2`.

All four modules were freshly rechecked together for this extension, with unchanged source hashes and zero warnings. The individual logs and declaration-by-declaration axiom comparisons are retained in `extension/review/navier-stokes/local-kernel-verification.json`. Thus the companion contains 82 selected declarations: 25 original interfaces, seven quantum interfaces, sixteen diagram interfaces and these 34 fluid interfaces. This is a count of exact declarations, not a claim that 82 complete manuscript theorems, every numerical implementation, or the continuum singularity construction have been formalized here.

Family	Count	Checked content and its boundary
Finite energy	11	Finite real matrix work identities, pressure cancellation under the stated constraint, nonnegative factorized viscosity, and an actual time derivative of coordinate kinetic energy. No continuum regularity conclusion.

Family	Count	Checked content and its boundary
Orthogonal reduction	5	Kernel residual, orthogonality, Pythagorean energy split, energy nonincrease and best approximation for a symmetric idempotent matrix. No closure or trajectory error theorem.
Chain continuity	2	Boundary flux has zero subsequent boundary, and adding such a flux preserves the divergence. The complex equation is explicit.
Schur elimination	1	Equivalence of a block system and its eliminated system when the required block is a linear equivalence. No assumption that every block is invertible.
Discrete memory	3	All-step hidden-state expansion, resolved memory identity and the obstruction to an instantaneous closure. No decay assumption is inserted into the recurrence.
Forbidden composition	1	A strict lower block, then a block diagonal map, then another strict lower block has zero composite. This is one exact operator pattern from the Volterra discussion.
Quadratic order	11	The faithful order representation, addition and multiplication, norm determinant, matrix polynomial, determinant of every power, and absence of a commuting real linear complex structure. The topological covering interpretation is proved separately in the manuscript.

The component map in `extension/review/navier-stokes/formalization-map.json` and the main claim register link these declarations to the statements they actually support. A matrix energy identity is a component of the finite ODE existence argument, not a Lean proof of that entire existence theorem. The exact discrete memory recurrence is not a Lean proof of the continuous integral formula. Likewise the determinant of J^k is not, by itself, a formalized covering-space degree theorem.

A.2 The separate upstream Navier–Stokes formalization

The released project uses a different toolchain: Lean 4.34.0-rc2 and its own pinned mathlib dependency. Its repository is pinned at

`f9e8bc5b38b6e212696e8a30e3e91517af887bbd`.

The audited final solution target has a local import closure of 609 modules. A comment- and string-aware source scan of this closure finds no admitted placeholders or project axiom declarations, and no challenge module is imported by that closure. This is useful source evidence but is not a kernel check.

A fresh upstream build was attempted with the exact verified compiler archive. Dependency-cache retrieval encountered HTTP 429 responses. The subsequent direct target check stopped because imported build artifacts were missing, before theorem-body elaboration. The full upstream kernel check and independent Comparator check therefore remain uncompleted in this work. The supplied `release-build-verification.json` records these outcomes, while `release-verify.py` supplies a pinned reproduction route. A failed dependency download neither verifies nor refutes the theorem. The local 34-declaration result is kept entirely separate from this upstream result.

The complete 166-page paper was read across the two mathematical reviewers, with every page attributed in `release-source-manifest.json`. That reading supports the exposition and the precise statement of the forced whole-space and periodic results. It does not substitute for the missing fresh formal build, nor does it turn a proof of forced breakdown into an unforced theorem.

A.3 Independent finite computational checks

The new program `extension/code/verify_fluid_interfaces.py` constructs exact integer cell complexes for six periodic square grids, with $L = 3, \ldots, 8$. It checks the zero boundary composite and the expected finite ranks. It independently evaluates symbolic Taylor–Green equations and dense complex-matrix circuit operations. Its machine-readable output is `extension/verification/fluids-python.json`.

The browser core is separately checked by `extension/site/verify-fluids.cjs`, which consumes independent Python reference data. The recorded comparison covers sixteen circuit inputs, 64 parameter-feature derivative cases and 8,865 complete toral-fibre points. The maximum amplitude discrepancy is about 2.22×10^{-16}; the maximum character-pullback discrepancy is about 1.01×10^{-13}; and the heat-channel probability discrepancy is about 3.47×10^{-18}. These are floating-point discrepancies, not exact-real proof bounds. The finite-difference momentum residual decreases from 3.03×10^{-6} to 7.57×10^{-7} to 1.89×10^{-7} under successive halvings of the step, consistent with the method's second-order error on this manufactured solution.

The channel checks distinguish the live diagonal-state simulation from the separate coherent-state matrix test. Positivity, trace preservation and complete positivity follow analytically from the Kraus form in Chapter 4; finite Choi and density-matrix calculations test the implemented instances. The circuit is simulated classically and is untrained. The circuit register and the periodic position register are distinct systems. No NVIDIA acceleration is required or claimed for these small new examples; the earlier GPU surface-code and ToricGT experiments retain their separate recorded execution evidence.

A.4 Reproduction commands

From the repository root, the finite computations can be repeated with:

```
python3 extension/code/verify_fluid_interfaces.py
node extension/site/verify-fluids.cjs
```

The first script requires the numerical and symbolic dependencies stated in its source. The second uses the generated reference data. The local formal check can be repeated using the installed pinned toolchain:

```
python3 extension/review/navier-stokes/formalization-verify.py \
  --toolchain /absolute/path/to/lean-4.24.0-linux \
  --mathlib /absolute/path/to/mathlib4-v4.24.0
```

Alternatively, from a correctly resolved Lake package, run

```
lake env lean FluidInterfaces.lean
```

The manuscript entry points share the same chapter sources. With the listed TeX packages installed:

```
python3 extension/manuscript/build_pdf.py
python3 extension/manuscript/build_pdf.py \
  --main fluid-extension --min-pages 1
```

The full monograph keeps the 100-page lower gate. The standalone supplement is not padded to a target length. Both builds reject undefined references, missing characters and overfull boxes, and retain their actual compiler logs and source hashes. Visual review is recorded separately from those mechanical gates.

A.5 The complete selected fluid formalization

The following is the source actually checked, including its type and axiom audit block. Its finite types are part of the theorem statements; replacing them with function spaces would create a different theorem and new analytic obligations.

```
import Mathlib.LinearAlgebra.Matrix.PosDef
import Mathlib.LinearAlgebra.Matrix.ToLin
import Mathlib.LinearAlgebra.Matrix.Notation
import Mathlib.LinearAlgebra.Matrix.Determinant.Basic
import Mathlib.Algebra.Module.LinearMap.End
import Mathlib.Analysis.Calculus.Deriv.Add
import Mathlib.Analysis.Calculus.Deriv.Mul
import Mathlib.Tactic.Ring
import Mathlib.Tactic.Linarith

/-!
# Finite fluid and information-reduction interfaces

The matrix results below concern actual finite real matrices, not a continuum
existence or regularity theorem. In particular, an energy identity for every
finite discretization supplies no mesh-uniform derivative or compactness bound.
Skew transport, positive factorized closures, orthogonal projection, and chain
complexes provide independently checkable structural constraints for a fluid
model on a surface cellulation. A learned closure may depend on the current
state: the corresponding matrix identities apply pointwise at that state.

The block-elimination and memory results describe exact finite linear reduction.
They do not identify its memory kernel with the nonlinear Navier--Stokes one,
and do not assert that an instantaneous closure is exact.

This file adds selected mathematical interfaces to the existing formalization.
It does not reprove or import the OpenAI Navier--Stokes release.
-/

set_option autoImplicit false

open Matrix
open scoped BigOperators Matrix

namespace FluidInterfaces

section Energy

variable {n m p : Type*} [Fintype n] [Fintype m] [Fintype p]

/-- Finite Euclidean kinetic energy with unit mass matrix. -/
noncomputable def kineticEnergy (u : n → ℝ) : ℝ := (u ⬝ᵥ u) / 2

theorem dot_self_nonneg (u : n → ℝ) : 0 ≤ u ⬝ᵥ u := by
  exact Finset.sum_nonneg (fun i _ => mul_self_nonneg (u i))

theorem kineticEnergy_nonneg (u : n → ℝ) : 0 ≤ kineticEnergy u :=
  div_nonneg (dot_self_nonneg u) (by norm_num)

/-- The transpose identity is valid also for rectangular finite matrices. -/
theorem transpose_work (L : Matrix m n ℝ) (u : n → ℝ) (v : m → ℝ) :
    u ⬝ᵥ (Lᵀ *ᵥ v) = (L *ᵥ u) ⬝ᵥ v := by
  rw [dotProduct_mulVec, vecMul_transpose]
```

```
/-- Actual matrix skew-symmetry implies exactly zero quadratic work. -/
theorem skew_transport_work (J : Matrix n n ℝ) (hJ : Jᵀ = -J)
    (u : n → ℝ) : u ⬝ᵥ (J *ᵥ u) = 0 := by
  have h := transpose_work J u u
  rw [hJ, neg_mulVec, dotProduct_neg, dotProduct_comm (J *ᵥ u) u] at h
  linarith

/-- A factorized dissipative closure has exactly squared-flux work. -/
theorem factorized_viscosity_work (L : Matrix m n ℝ) (u : n → ℝ) :
    u ⬝ᵥ ((Lᵀ * L) *ᵥ u) = (L *ᵥ u) ⬝ᵥ (L *ᵥ u) := by
  rw [← mulVec_mulVec, transpose_work]

theorem factorized_viscosity_nonneg (L : Matrix m n ℝ) (u : n → ℝ) :
    0 ≤ u ⬝ᵥ ((Lᵀ * L) *ᵥ u) := by
  rw [factorized_viscosity_work]
  exact dot_self_nonneg _

/-- Discrete gradient pressure does no work on a divergence-free velocity. -/
theorem pressure_work_zero (B : Matrix p n ℝ) (u : n → ℝ) (pressure : p → ℝ)
    (hu : B *ᵥ u = 0) : u ⬝ᵥ (Bᵀ *ᵥ pressure) = 0 := by
  rw [transpose_work, hu, zero_dotProduct]

/-- Exact finite energy-work balance including pressure, physical viscosity,
and a separately factorized learned closure. No continuum limit is asserted. -/
theorem finite_fluid_work_identity
    (J : Matrix n n ℝ) (hJ : Jᵀ = -J)
    (D : Matrix m n ℝ) (L : Matrix p n ℝ) (ν : ℝ)
    (u force : n → ℝ) :
    u ⬝ᵥ (J *ᵥ u - ν • ((Dᵀ * D) *ᵥ u) - (Lᵀ * L) *ᵥ u + force) =
      u ⬝ᵥ force - ν * ((D *ᵥ u) ⬝ᵥ (D *ᵥ u)) -
        (L *ᵥ u) ⬝ᵥ (L *ᵥ u) := by
  rw [dotProduct_add, dotProduct_sub, dotProduct_sub,
    skew_transport_work J hJ, dotProduct_smul,
    factorized_viscosity_work, factorized_viscosity_work]
  simp only [smul_eq_mul]
  ring

theorem unforced_work_nonpositive
    (J : Matrix n n ℝ) (hJ : Jᵀ = -J)
    (D : Matrix m n ℝ) (L : Matrix p n ℝ) (ν : ℝ) (hν : 0 ≤ ν)
    (u : n → ℝ) :
    u ⬝ᵥ (J *ᵥ u - ν • ((Dᵀ * D) *ᵥ u) - (Lᵀ * L) *ᵥ u) ≤ 0 := by
  have h := finite_fluid_work_identity J hJ D L ν u 0
  simp only [add_zero, dotProduct_zero] at h
  rw [h]
  have hv := mul_nonneg hν (dot_self_nonneg (D *ᵥ u))
  have hl := dot_self_nonneg (L *ᵥ u)
  linarith

/-- This is an actual time derivative, proved from the coordinate derivatives. -/
theorem kinetic_energy_derivative (u : ℝ → n → ℝ) (v : n → ℝ) (t : ℝ)
    (hu : ∀ i, HasDerivAt (fun s => u s i) (v i) t) :
    HasDerivAt (fun s => kineticEnergy (u s)) ((u t) ⬝ᵥ v) t := by
  have hi : ∀ i, HasDerivAt (fun s => u s i * u s i / 2) (u t i * v i) t := by
    intro i
    convert ((hu i).mul (hu i)).div_const 2 using 1; ring
  have hs := HasDerivAt.fun_sum (u := Finset.univ) (fun i _ => hi i)
  simpa only [kineticEnergy, dotProduct, Finset.sum_div] using hs

/-- The exact semidiscrete ODE yields a genuine differential energy identity.
It remains a finite model: no uniform continuum estimate is implicit. -/
theorem finite_fluid_energy_derivative
    (J : Matrix n n ℝ) (hJ : Jᵀ = -J)
    (D : Matrix m n ℝ) (L : Matrix p n ℝ) (ν : ℝ)
    (u : ℝ → n → ℝ) (force : n → ℝ) (t : ℝ)
    (hu : ∀ i, HasDerivAt (fun s => u s i)
```

```
      ((J *ᵥ u t - ν • ((Dᵀ * D) *ᵥ u t) - (Lᵀ * L) *ᵥ u t + force) i) t) :
    HasDerivAt (fun s => kineticEnergy (u s))
      (u t ⬝ᵥ force - ν * ((D *ᵥ u t) ⬝ᵥ (D *ᵥ u t)) -
        (L *ᵥ u t) ⬝ᵥ (L *ᵥ u t)) t := by
  have h := kinetic_energy_derivative u _ t hu
  rwa [finite_fluid_work_identity J hJ D L ν (u t) force] at h

end Energy

section OrthogonalReduction

variable {n : Type*} [Fintype n]

/-- Idempotence sends the discarded part into the projector's kernel. -/
theorem projection_residual_kernel (P : Matrix n n ℝ) (hP : P * P = P)
    (u : n → ℝ) : P *ᵥ (u - P *ᵥ u) = 0 := by
  rw [mulVec_sub, mulVec_mulVec, hP, sub_self]

/-- Symmetry and idempotence imply orthogonality of retained and discarded data. -/
theorem projection_residual_orthogonal (P : Matrix n n ℝ)
    (hs : Pᵀ = P) (hi : P * P = P) (u v : n → ℝ) :
    (u - P *ᵥ u) ⬝ᵥ (P *ᵥ v) = 0 := by
  rw [← hs, transpose_work, hs, projection_residual_kernel P hi, zero_dotProduct]

/-- Exact information loss of an orthogonal linear reduction in kinetic energy. -/
theorem projection_energy_split (P : Matrix n n ℝ)
    (hs : Pᵀ = P) (hi : P * P = P) (u : n → ℝ) :
    kineticEnergy u = kineticEnergy (P *ᵥ u) + kineticEnergy (u - P *ᵥ u) := by
  have ho := projection_residual_orthogonal P hs hi u u
  simp only [sub_dotProduct] at ho
  have hc := dotProduct_comm u (P *ᵥ u)
  simp only [kineticEnergy, sub_dotProduct, dotProduct_sub]
  linarith

theorem projection_energy_nonincrease (P : Matrix n n ℝ)
    (hs : Pᵀ = P) (hi : P * P = P) (u : n → ℝ) :
    kineticEnergy (P *ᵥ u) ≤ kineticEnergy u := by
  have hsplit := projection_energy_split P hs hi u
  have hnonneg := kineticEnergy_nonneg (u - P *ᵥ u)
  linarith

/-- The same split proves optimality against every vector in the retained range. -/
theorem projection_best_approximation (P : Matrix n n ℝ)
    (hs : Pᵀ = P) (hi : P * P = P) (u v : n → ℝ) (hv : P *ᵥ v = v) :
    kineticEnergy (u - P *ᵥ u) ≤ kineticEnergy (u - v) := by
  have hsplit := projection_energy_split P hs hi (u - v)
  have hp : P *ᵥ (u - v) = P *ᵥ u - v := by rw [mulVec_sub, hv]
  have hr : u - v - (P *ᵥ u - v) = u - P *ᵥ u := by abel
  rw [hp, hr] at hsplit
  rw [hsplit]
  exact le_add_of_nonneg_left (kineticEnergy_nonneg _)

end OrthogonalReduction

section Continuity

variable {n m p : Type*} [Fintype n] [Fintype p]

/-- A flux represented by the next boundary automatically preserves divergence. -/
theorem boundary_flux_divergence_zero (B : Matrix m n ℝ) (C : Matrix n p ℝ)
    (hcomplex : B * C = 0) (q : p → ℝ) : B *ᵥ (C *ᵥ q) = 0 := by
  rw [mulVec_mulVec, hcomplex, zero_mulVec]

/-- The exact algebraic continuity constraint survives every such flux update. -/
theorem boundary_flux_update_preserves_divergence
    (B : Matrix m n ℝ) (C : Matrix n p ℝ) (hcomplex : B * C = 0)
    (u : n → ℝ) (q : p → ℝ) :
    B *ᵥ (u + C *ᵥ q) = B *ᵥ u := by
```

```
  rw [mulVec_add, boundary_flux_divergence_zero B C hcomplex, add_zero]

end Continuity

section BlockElimination

variable {K X Y : Type*} [Field K]
variable [AddCommGroup X] [Module K X] [AddCommGroup Y] [Module K Y]

/-- The exact reduced block; invertibility is an explicit analytic obstruction. -/
def schurComplement (A : X →ₗ[K] X) (B : Y →ₗ[K] X)
    (C : X →ₗ[K] Y) (D : Y ≃ₗ[K] Y) : X →ₗ[K] X :=
  A - B.comp (D.symm.toLinearMap.comp C)

/-- Eliminating the hidden variable is equivalent to the original block system,
not merely a one-way necessary condition. -/
theorem schur_elimination_iff
    (A : X →ₗ[K] X) (B : Y →ₗ[K] X)
    (C : X →ₗ[K] Y) (D : Y ≃ₗ[K] Y)
    (x : X) (y : Y) (f : X) (g : Y) :
    (A x + B y = f ∧ C x + D y = g) ↔
      (schurComplement A B C D x = f - B (D.symm g) ∧
        y = D.symm (g - C x)) := by
  have hy : C x + D y = g ↔ y = D.symm (g - C x) := by
    constructor
    · intro h
      have hd : D y = g - C x := by rw [← h]; abel
      simpa only [D.symm_apply_apply] using congrArg D.symm hd
    · intro h
      rw [h, D.apply_symm_apply]
      abel
  rw [hy]
  constructor
  · rintro ⟨hf, hy'⟩
    refine ⟨?_, hy'⟩
    rw [hy', map_sub, map_sub] at hf
    simp only [schurComplement, LinearMap.sub_apply, LinearMap.comp_apply,
      LinearEquiv.coe_coe]
    rw [← hf]
    abel
  · rintro ⟨hf, hy'⟩
    refine ⟨?_, hy'⟩
    simp only [schurComplement, LinearMap.sub_apply, LinearMap.comp_apply,
      LinearEquiv.coe_coe] at hf
    rw [hy', map_sub, map_sub]
    rw [sub_eq_iff_eq_add] at hf
    rw [hf]
    abel

end BlockElimination

section ExactMemory

variable {K X Y : Type*} [Field K]
variable [AddCommGroup X] [Module K X] [AddCommGroup Y] [Module K Y]

/-- Eliminating a hidden linear recurrence produces its entire past dependence. -/
theorem hidden_recurrence_expansion (C : X →ₗ[K] Y) (D : Module.End K Y)
    (p : ℕ → X) (q : ℕ → Y)
    (hstep : ∀ n : ℕ, q (n + 1) = D (q n) + C (p n)) (n : ℕ) :
    q n = (D ^ n) (q 0) +
      ∑ k ∈ Finset.range n, (D ^ (n - 1 - k)) (C (p k)) := by
  induction n with
  | zero => simp
  | succ n ih =>
    rw [hstep, ih, map_add, map_sum, Finset.sum_range_succ]
    simp only [Nat.add_sub_cancel, Nat.sub_self, pow_zero, Module.End.one_apply]
    have hsum :
```

```
      (∑ k ∈ Finset.range n, D ((D ^ (n - 1 - k)) (C (p k)))) =
        ∑ k ∈ Finset.range n, (D ^ (n - k)) (C (p k)) := by
    apply Finset.sum_congr rfl
    intro k hk
    have hklt : k < n := Finset.mem_range.mp hk
    have he : n - k = (n - 1 - k) + 1 := by omega
    rw [he, pow_succ', Module.End.mul_apply]
  rw [hsum, pow_succ', Module.End.mul_apply]
  abel

/-- The exact reduced dynamics contains a forcing from the hidden initial state
and the finite convolution with kernel B D^j C. -/
theorem resolved_memory_identity (A : Module.End K X) (B : Y →₁[K] X)
    (C : X →₁[K] Y) (D : Module.End K Y) (p : ℕ → X) (q : ℕ → Y)
    (hp : ∀ n : ℕ, p (n + 1) = A (p n) + B (q n))
    (hq : ∀ n : ℕ, q (n + 1) = D (q n) + C (p n)) (n : ℕ) :
    p (n + 1) = A (p n) + B ((D ^ n) (q 0)) +
      ∑ k ∈ Finset.range n, B ((D ^ (n - 1 - k)) (C (p k))) := by
  rw [hp, hidden_recurrence_expansion C D p q hq n, map_add, map_sum]
  abel

/-- There need be no exact autonomous deterministic closure on retained data:
two states with the same retained coordinates can have distinct next states. -/
theorem hidden_state_prevents_autonomous_closure
    (A : Module.End K X) (B : Y →₁[K] X) (y : Y) (hy : B y ≠ 0) :
    ¬ ∃ F : X → X, ∀ (x : X) (z : Y), F x = A x + B z := by
  rintro ⟨F, hF⟩
  have h₀ := hF 0 0
  have hy' := hF 0 y
  simp only [map_zero, add_zero, zero_add] at h₀ hy'
  exact hy (hy'.symm.trans h₀)

end ExactMemory

section NilpotentBlocks

variable {K X Y : Type*} [Field K]
variable [AddCommGroup X] [Module K X] [AddCommGroup Y] [Module K Y]

/-- A strict lower rectangular block on the direct sum of retained spaces. -/
def lowerBlock (L : X →₁[K] Y) : Module.End K (X × Y) where
  toFun xy := (0, L xy.1)
  map_add' a b := by simp
  map_smul' c a := by simp

/-- Any diagonal block between two strict lower blocks leaves zero composite.
The left and right lower blocks may differ, covering a parameter derivative. -/
theorem lower_diagonal_lower_zero (L M : X →₁[K] Y)
    (DX : Module.End K X) (DY : Module.End K Y) :
    (lowerBlock L).comp ((DX.prodMap DY).comp (lowerBlock M)) = 0 := by
  apply LinearMap.ext
  intro xy
  change (0, L (DX 0)) = (0, 0)
  simp

end NilpotentBlocks

section QuadraticOrder

/-- The integer matrix for multiplication by sqrt(2), in a conjugate basis. -/
def quadraticGenerator : Matrix (Fin 2) (Fin 2) ℤ := !![-1, 1; 1, 1]

/-- The degree-fourteen matrix appearing in the reviewed proof's toral stage. -/
def renormalizationMatrix : Matrix (Fin 2) (Fin 2) ℤ := !![3, 1; 1, 5]

/-- An exact faithful representation of the underlying quadratic order. -/
def quadraticOrderMatrix (a b : ℤ) : Matrix (Fin 2) (Fin 2) ℤ :=
  !![a - b, b; b, a + b]
```

```
theorem renormalization_is_quadratic_order_element :
    renormalizationMatrix = quadraticOrderMatrix 4 1 := by
  decide

theorem quadratic_order_one :
    quadraticOrderMatrix 1 0 = (1 : Matrix (Fin 2) (Fin 2) ℤ) := by
  decide

theorem quadratic_order_addition (a b c d : ℤ) :
    quadraticOrderMatrix a b + quadraticOrderMatrix c d =
      quadraticOrderMatrix (a + c) (b + d) := by
  ext i j
  fin_cases i <;> fin_cases j <;> simp [quadraticOrderMatrix] <;> ring

theorem quadratic_generator_square : quadraticGenerator * quadraticGenerator =
    (2 : ℤ) • (1 : Matrix (Fin 2) (Fin 2) ℤ) := by
  decide

theorem renormalization_quadratic_identity :
    renormalizationMatrix * renormalizationMatrix - 8 • renormalizationMatrix +
      (14 : ℤ) • (1 : Matrix (Fin 2) (Fin 2) ℤ) = 0 := by
  decide

theorem renormalization_determinant : Matrix.det renormalizationMatrix = 14 := by
  norm_num [renormalizationMatrix, Matrix.det_fin_two]

theorem renormalization_power_determinant (k : ℕ) :
    Matrix.det (renormalizationMatrix ^ k) = 14 ^ k := by
  rw [Matrix.det_pow, renormalization_determinant]

theorem quadratic_order_multiplication (a b c d : ℤ) :
    quadraticOrderMatrix a b * quadraticOrderMatrix c d =
      quadraticOrderMatrix (a * c + 2 * b * d) (a * d + b * c) := by
  ext i j
  fin_cases i <;> fin_cases j <;>
    simp [quadraticOrderMatrix, Matrix.mul_apply, Fin.sum_univ_two] <;> ring

theorem quadratic_order_determinant (a b : ℤ) :
    Matrix.det (quadraticOrderMatrix a b) = a ^ 2 - 2 * b ^ 2 := by
  simp [quadraticOrderMatrix, Matrix.det_fin_two]
  ring

theorem quadratic_order_representation_injective :
    Function.Injective (fun ab : ℤ × ℤ => quadraticOrderMatrix ab.1 ab.2) := by
  intro ab cd h
  have h₀ := congrArg (fun M : Matrix (Fin 2) (Fin 2) ℤ => M 0 0) h
  have h₁ := congrArg (fun M : Matrix (Fin 2) (Fin 2) ℤ => M 0 1) h
  change ab.1 - ab.2 = cd.1 - cd.2 at h₀
  change ab.2 = cd.2 at h₁
  apply Prod.ext
  · linarith
  · exact h₁

/-- This real toral endomorphism cannot be complex-linear for any real linear
complex structure: no commuting real matrix squares to minus the identity. -/
theorem no_commuting_complex_structure (I : Matrix (Fin 2) (Fin 2) ℝ)
    (hc : I * !![3, 1; 1, 5] = !![3, 1; 1, 5] * I) :
    I * I ≠ -(1 : Matrix (Fin 2) (Fin 2) ℝ) := by
  intro hsq
  have hc₀ := congrArg (fun M : Matrix (Fin 2) (Fin 2) ℝ => M 0 0) hc
  have hs₀ := congrArg (fun M : Matrix (Fin 2) (Fin 2) ℝ => M 0 0) hsq
  simp [Matrix.mul_apply, Fin.sum_univ_two] at hc₀ hs₀
  have hb : I 0 1 = I 1 0 := by linarith
  rw [← hb] at hs₀
  nlinarith [sq_nonneg (I 0 0), sq_nonneg (I 0 1)]

end QuadraticOrder
```

```
end FluidInterfaces

/-! Exact theorem types and trusted-kernel foundational dependency audit. -/
#check FluidInterfaces.renormalization_is_quadratic_order_element
#check FluidInterfaces.quadratic_order_one
#check FluidInterfaces.quadratic_order_addition
#check FluidInterfaces.dot_self_nonneg
#check FluidInterfaces.kineticEnergy_nonneg
#check FluidInterfaces.transpose_work
#check FluidInterfaces.skew_transport_work
#check FluidInterfaces.factorized_viscosity_work
#check FluidInterfaces.factorized_viscosity_nonneg
#check FluidInterfaces.pressure_work_zero
#check FluidInterfaces.finite_fluid_work_identity
#check FluidInterfaces.unforced_work_nonpositive
#check FluidInterfaces.kinetic_energy_derivative
#check FluidInterfaces.finite_fluid_energy_derivative
#check FluidInterfaces.projection_residual_kernel
#check FluidInterfaces.projection_residual_orthogonal
#check FluidInterfaces.projection_energy_split
#check FluidInterfaces.projection_energy_nonincrease
#check FluidInterfaces.projection_best_approximation
#check FluidInterfaces.boundary_flux_divergence_zero
#check FluidInterfaces.boundary_flux_update_preserves_divergence
#check FluidInterfaces.schur_elimination_iff
#check FluidInterfaces.hidden_recurrence_expansion
#check FluidInterfaces.resolved_memory_identity
#check FluidInterfaces.hidden_state_prevents_autonomous_closure
#check FluidInterfaces.lower_diagonal_lower_zero
#check FluidInterfaces.quadratic_generator_square
#check FluidInterfaces.renormalization_quadratic_identity
#check FluidInterfaces.renormalization_determinant
#check FluidInterfaces.renormalization_power_determinant
#check FluidInterfaces.quadratic_order_multiplication
#check FluidInterfaces.quadratic_order_determinant
#check FluidInterfaces.quadratic_order_representation_injective
#check FluidInterfaces.no_commuting_complex_structure

#print axioms FluidInterfaces.renormalization_is_quadratic_order_element
#print axioms FluidInterfaces.quadratic_order_one
#print axioms FluidInterfaces.quadratic_order_addition
#print axioms FluidInterfaces.dot_self_nonneg
#print axioms FluidInterfaces.kineticEnergy_nonneg
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#print axioms FluidInterfaces.skew_transport_work
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#print axioms FluidInterfaces.kinetic_energy_derivative
#print axioms FluidInterfaces.finite_fluid_energy_derivative
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#print axioms FluidInterfaces.projection_residual_orthogonal
#print axioms FluidInterfaces.projection_energy_split
#print axioms FluidInterfaces.projection_energy_nonincrease
#print axioms FluidInterfaces.projection_best_approximation
#print axioms FluidInterfaces.boundary_flux_divergence_zero
#print axioms FluidInterfaces.boundary_flux_update_preserves_divergence
#print axioms FluidInterfaces.schur_elimination_iff
#print axioms FluidInterfaces.hidden_recurrence_expansion
#print axioms FluidInterfaces.resolved_memory_identity
#print axioms FluidInterfaces.hidden_state_prevents_autonomous_closure
#print axioms FluidInterfaces.lower_diagonal_lower_zero
#print axioms FluidInterfaces.quadratic_generator_square
#print axioms FluidInterfaces.renormalization_quadratic_identity
#print axioms FluidInterfaces.renormalization_determinant
```

```
#print axioms FluidInterfaces.renormalization_power_determinant
#print axioms FluidInterfaces.quadratic_order_multiplication
#print axioms FluidInterfaces.quadratic_order_determinant
#print axioms FluidInterfaces.quadratic_order_representation_injective
#print axioms FluidInterfaces.no_commuting_complex_structure
```

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